



# **C H A P T E R 12B**

## ***Electronic Data Interchange (EDI)***

- 12.17 Introduction to Electronic Data Interchange (EDI)*
- 12.18 Definition of EDI*
- 12.19 Network-enabled Business Practices*
- 12.20 Applications of EDI*
- 12.21 E-commerce Development Trends in 2018*
- 12.22 Advantages of EDI*
- 12.23 Limitations of EDI*
- 12.24 Concept of EDI*
- 12.25 Disadvantages of EDI*
- 12.26 Cost of EDI*
- 12.27 EDI Model*
- 12.28 Protocol*
- 12.29 Data Standards Used in EDI*

## 12.17 INTRODUCTION TO ELECTRONIC DATA INTERCHANGE (EDI)

Electronic Data Interchange (EDI) is the direct computer-to-computer exchange between two organisations of standard business transaction documents, such as invoices, bill of lading and purchase orders (in a standard format). It saves money and time because transactions can be transmitted from one information system to another through a telecommunications network, eliminating the printing and handling of paper at one end and the inputting of data at the other. The formats look like standard forms and highly structured. One widely used format is for purchase orders and consists of an outer digital 'envelope' with the address of both the sender and the receiver. Any firm can establish new relationship with its suppliers. EDI provides strategic benefits by helping a firm 'lock-in' customers, making it easier for customers or distributors.

The working with and without EDI is as shown in Figure 20.19.

### Purchaser

Department orders for an item to the purchasing department. Purchasing department prepares the purchase order and sends it to the vendor via its mail office under copies to accounts and shipping departments. Computers are involved everywhere but they are used for their own limited purposes.

### Vendor

Vendor receives the PO from its mail office, route it to its sales. Sale department will take action through shipping and deliver the goods to the receiving department of the purchaser. The copies of order confirmation bill will be delivered to the mail office of the purchases. Computers are involved but they are used for their own purpose.

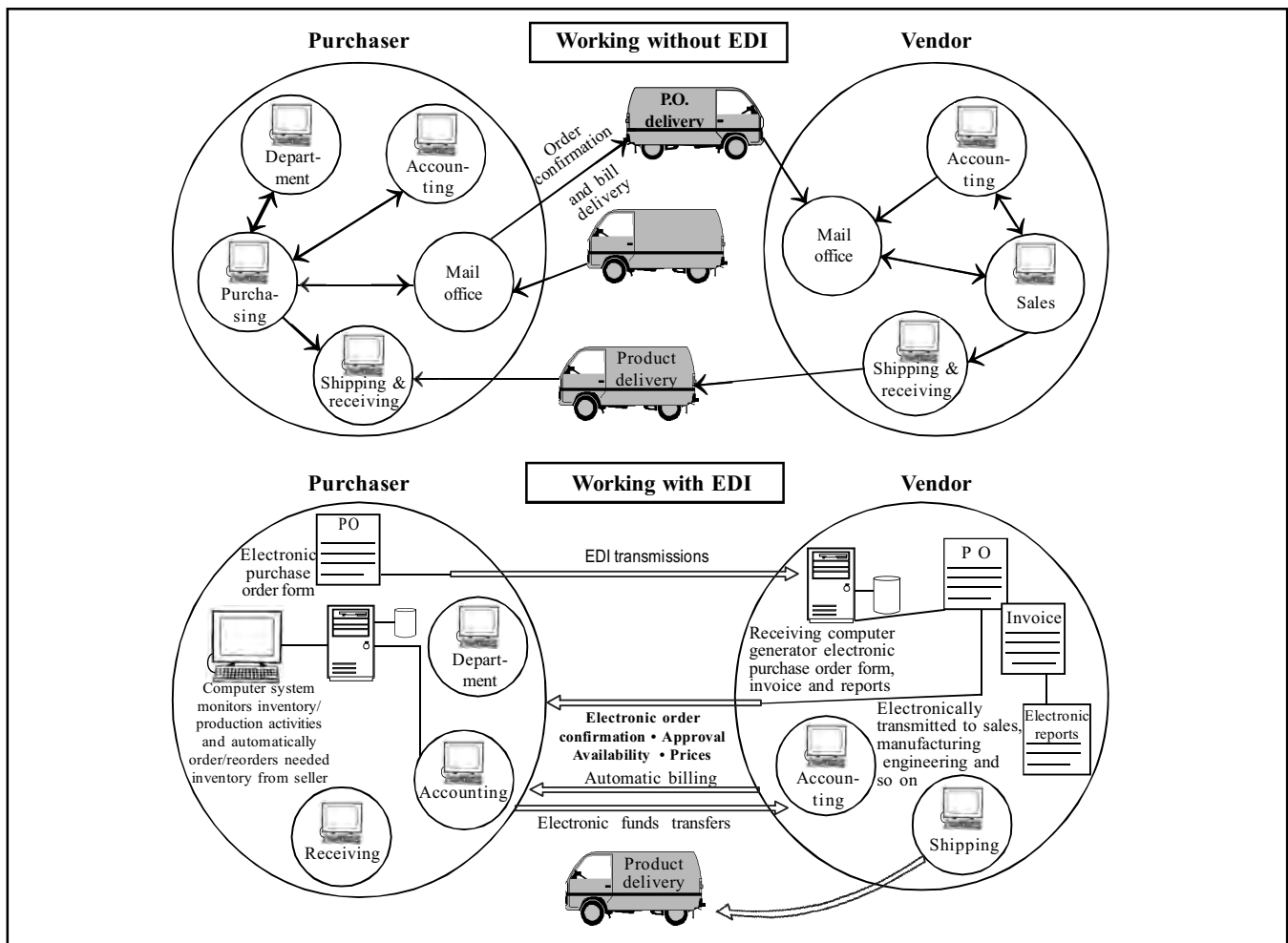
EDI differs from electronic mail in that it transmits an actual structured transaction (fields like transaction date/amount, senders name, recipient's name, etc.) in contrast to an unstructured text message such as a letter. EDI can also curb inventory costs by minimising the amount of time, components are in inventory.

Organisation can most benefit from EDI when they integrate the data supplied by EDI with applications such as accounts payable, inventory control, shipping and production planning.

To make EDI to work properly, four key requirements exists. These are:

- Transaction standardisation (Transaction formats and data must be standardised).
- Transaction software (Special software to be developed to convert incoming and outgoing messages into a form suitable to other companies).
- Appropriate 'mail box' facilities (Companies using EDI must select a third party, value-added network with mail box facilities that allow messages to be sent, sorted and held until they are needed by the receiving computer).
- Legal restrictions (To comply with legal requirements, certain transactions require "writing" or the "original document" in hard copy form).

Direct computer-to-computer exchange between two organisations of standard business transaction documents are known as EDI. The documents may be such as invoices, bill of lading, or purchase orders. The main purpose of EDI is to save money and time because transactions can be transmitted from one information system to another through a telecommunications network, eliminating the printing and handling of papers at one end and the inputting of data at the other. It may also provide strategic benefits by helping a firm 'lock-in' customers, making it easier for customers or distributors to order from them rather than from competitors. It differs from electronic mail in that it transmits an actual structured transaction (with fields such as the transaction date, transaction amount, sender's name and recipients name) as opposed to unstructured text message such as a letter.



**Fig. 12.19 – An illustration of how EDI would improve and benefit business relationships among organisations**

**Note :** Here, lot of movements of various papers are done as shown in the centre. This can be avoided by computer-to-computer exchange in EDI. Please note that in the case of working with EDI, physical movements of the papers are avoided and time per each movement can be reduced since all these activities are computer-to-computer exchange.

Company uses EDI to automate price, shipping, receiving, and payment transactions with its customers as an example. Price updates and shipping notices are entered by the appropriate departments directly into company's computer system, which then transmits to the customer's computer systems. Similarly, customers material releases, receiving reports, and payment data are also transmitted directly through the computer systems back to the company. EDI has replaced paper for these transactions.

It lowers:

- Routine transaction processing cost
- Turnaround time (Less need to transfer data from hard copy forms into computer-ready transactions)
- Transcription errors
- Associated costs occurring during entry of data and printed out many times
- Inventory costs (by minimising the amount of time components are in inventory)

It benefits the organisation when the latter integrates the data supplied by EDI with applications such as accounts payable, inventory control, shipping and production planning.

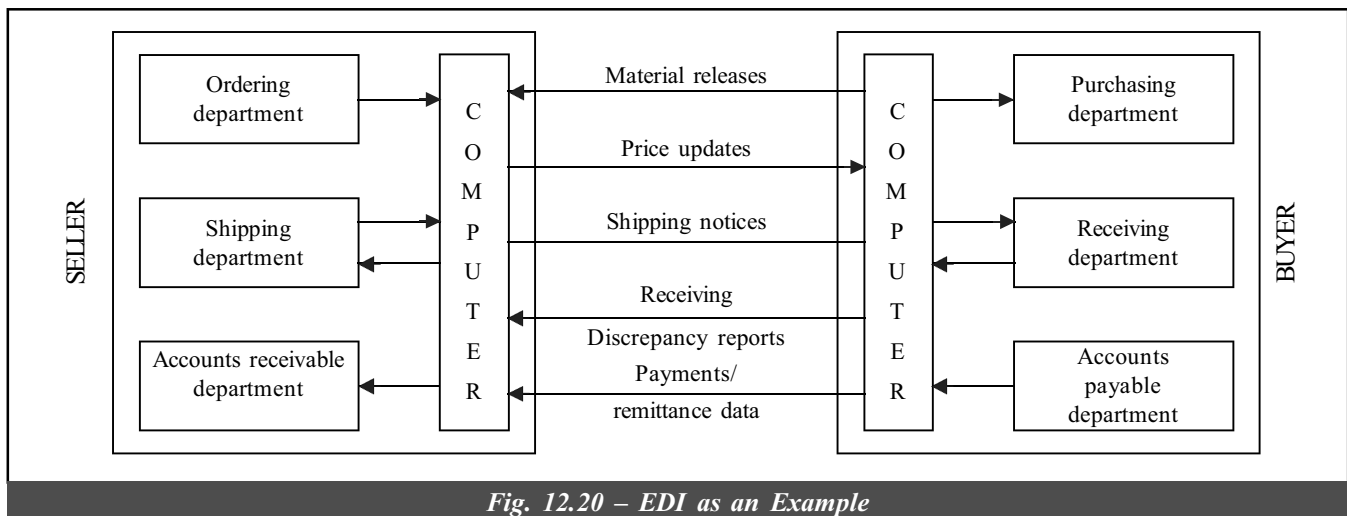


Fig. 12.20 – EDI as an Example

Subsequently transfer data items to the next destination, a process called store and forward. Messages are collected at each switching centre where they are organised (grouped into batches) in a manner that makes efficient use of the data paths connected to that centre. An analogy can be drawn between message switching and the transmission of mail through the postal system. Compared with circuit switching, message switching can provide a substantial increase in communication link utilisation.

### 12.18 DEFINITION OF EDI

EDI is the controlled transfer of data between business and organisations via established security standards. EDI consists of standardised electronic message formats, for business documents such as requests for quotations, purchase orders, purchase change orders, bills of lading, receiving advices and invoices. EDI (the standard in electronic commerce) is not a choice. It is the inevitable way, business will be done. Today's banking and finance operations have long been dependent on EDI and EFT to ensure that money and securities are transferred to the company or individual that requested the action.

### 12.19 NETWORK-ENABLED BUSINESS PRACTICES

Network-enabled business practices are carried out in various areas.

- (i) Engineering – Technical data interchange (engineering)
- (ii) Mass – customisation (demand-driven manufacturing)
- (iii) Virtual and team-based enterprises
- (iv) Outsourcing and co-ordination of logistics
- (v) Desktop video conferencing
- (vi) Document work flow systems
- (vii) Electronic mail (communications)
- (viii) Electronic data interchange (procurement)

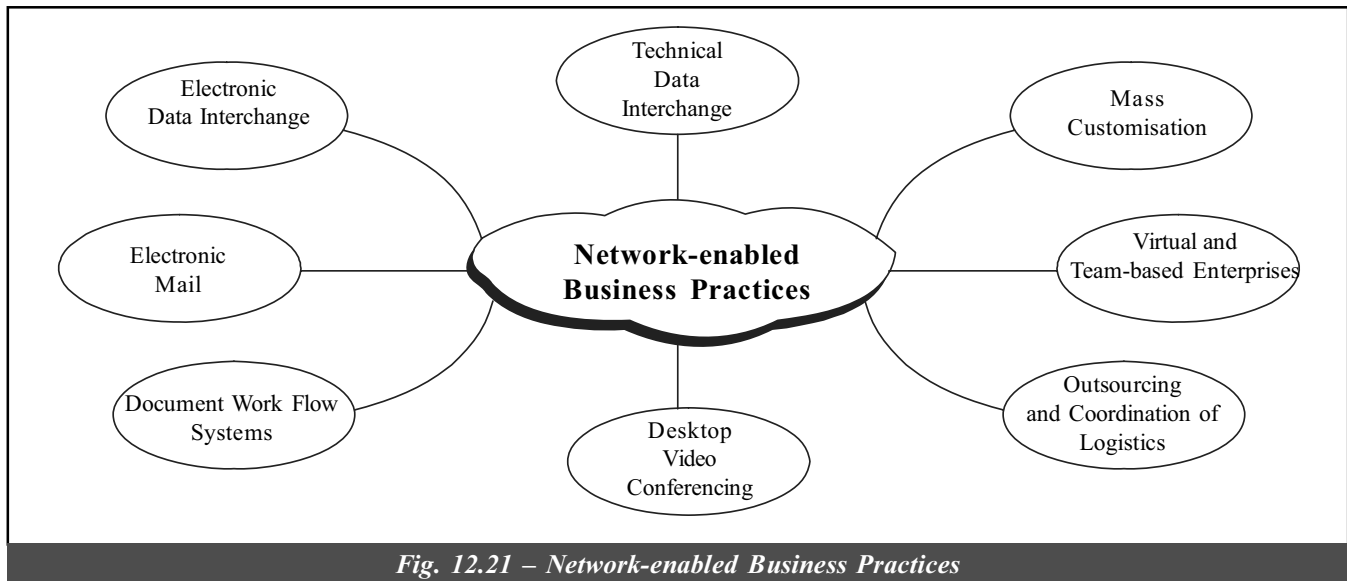


Fig. 12.21 – Network-enabled Business Practices

Electronic messaging technologies, during late 1980s and early 1990s, became an integral part of work flow or collaborative computing systems (known as groupware).

## 12.20 APPLICATIONS OF EDI

EDI is used in manufacturing, shipping, warehousing, utilities, pharmaceuticals, construction, petroleum, metals, banking, insurance, retailing, government, healthcare, and textiles among others.

## 12.21 E-COMMERCE DEVELOPMENT TRENDS IN 2018

E-commerce market is booming year by year. Nowadays, many people are making purchases from various e-commerce websites. According to stats by Statista, online shopping in the US alone has generated \$ 392.5 billion in the year 2017 and this figure is expected to increase by \$ 400 billion per year in 2018.

With rising competition, business owners want to make their website more attractive yet functional for their users. The latest development trends and technologies in e-commerce have showed developers to integrate various new technologies in their e-commerce website. EDI is one among them. By using the latest technology trends, e-commerce development companies want to improve the user experience of online shopping drastically. It is impossible to see that we are seeing currently in the development field a couple of years ago. Here are some of the top e-commerce development trends that business should follow in 2018.

**EDI and e-commerce** solutions designed are to enable companies to communicate business data electronically whether you are a small company looking for simple web-based EDI solution or a larger corporation searching for a reliable EDI system, scalable **cloud portal** can be used to support the company activities.

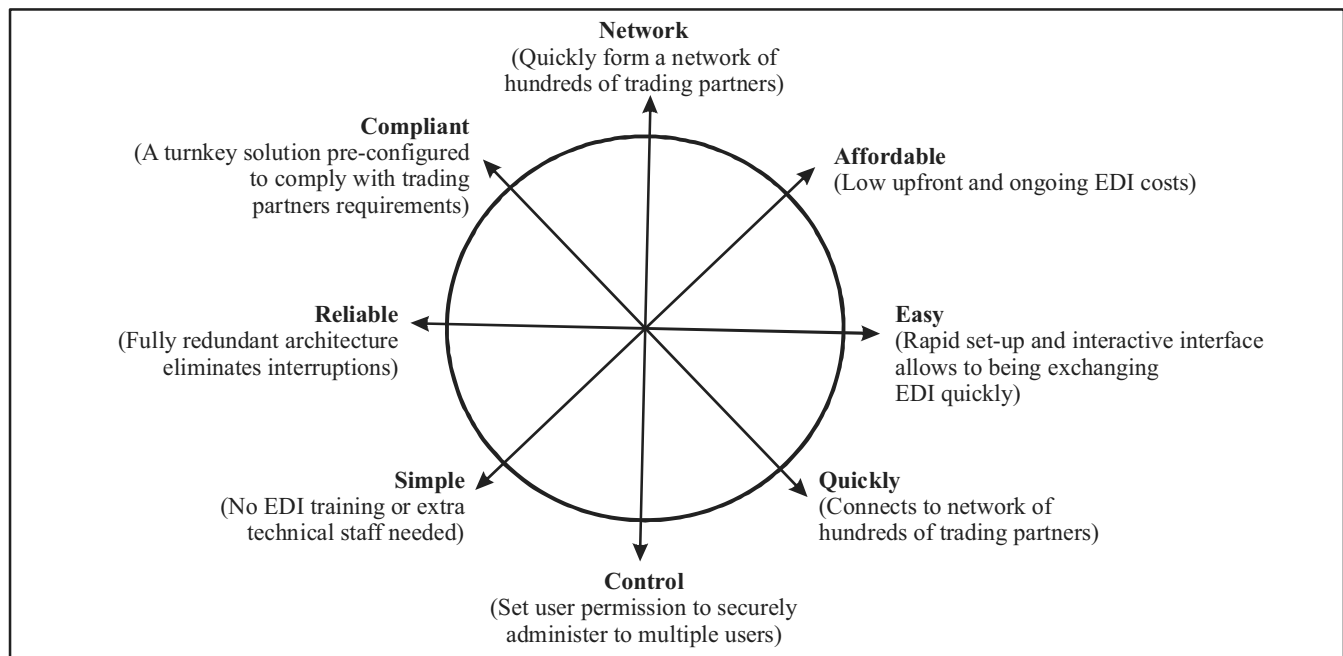
### Cloud-based (Web-based) EDI

Web EDI provides users with a simple and affordable way to become EDI capable. Streamline the procurement process with web EDI by easily managing transactions, processing and preparing orders and efficiently fulfilling shipments.

Cloud-based infrastructure provides maximum up-time so that EDI solution is always available to access. Web EDI requires no special software or computer knowledge. The only requirements is to have access to the internet. One can become EDI capable quickly with minimum cost and little training. Web EDI allows users to manage all of their EDI transactions from one simple, easy-to-use webpage. For e-mail users, web EDI is convenient.

### Web EDI Benefits

- Web EDI provides users with a simple and affordable way to become EDI capable. With a simple cloud portal, one can manage several activities.
- Order fulfillment can be made easy.



*Fig. 12.22 – Benefits with Web EDI*

#### Example (Data Trans at US)

(with no automated system or with queues and 24/7 monitoring provision)

#### Example:

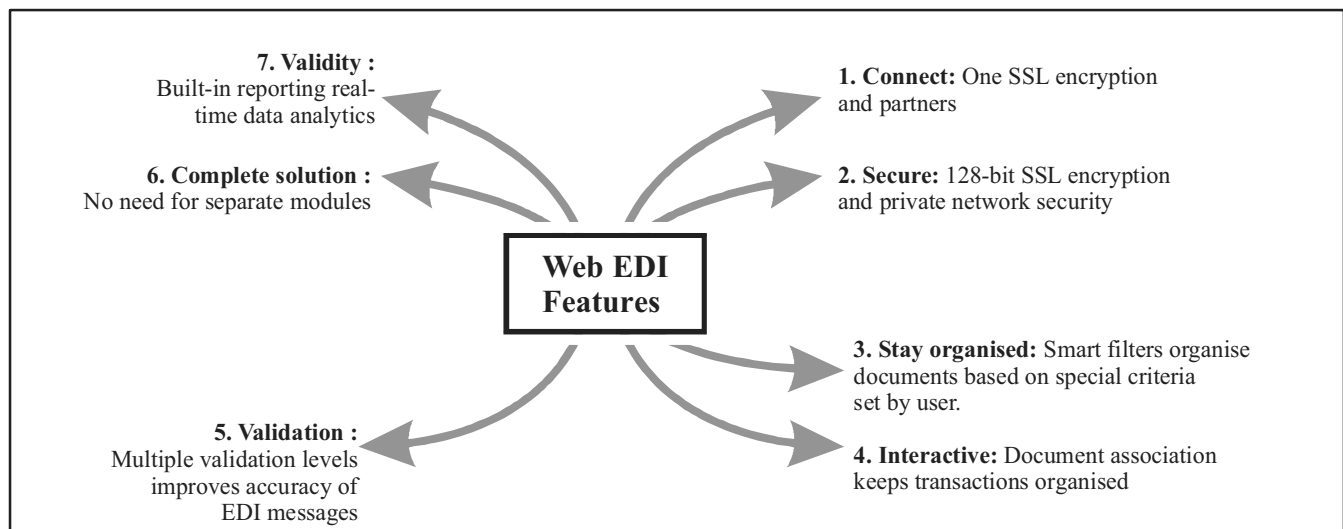
**Data Trans:** Is for cloud-based solutions, any trading partner any industry and any application.

This streamlines business processes with a feature-rich portal to enhance efficiency. An all-in-one solution that easily integrates with one's internal business processes by connecting the web EDI (cloud based) with:

- Shipping services such as ship station, FedEx or UPS to sync tracking numbers, generate shipping labels and fulfilling orders within seconds.
- Integrate with QuickBooks or any ERP to automate order and invoice processing.

#### Web EDI Features

Web EDI is the most complete EDI solution with features designed to make EDI as simple and easy as e-mail.



*Fig. 12.23 – Features of Web EDI*

## 12.22 ADVANTAGES OF EDI

EDI benefits are many. The major benefit is cost reduction by eliminating paper document handling and with faster electronic document transmission.

The other benefits are:

- Improvements in overall quality – By better record keeping, fewer errors in data, reduction in processing time, less reliance on human interpretation of data and minimised unproductive time.
- Inventory reduction. It permits faster and more accurate filling of orders, helps reduce inventory and assists in JIT (Just in Time) inventory management.
- Provides better information for management decision-making. It provides accurate information and audit trails for transactions, enabling business to identify areas offering the greatest potential for efficiency improvement or cost reduction.

EDI is used for sending invoices, purchase orders, custom documents, shipping notices and other types of business documents in a fast and expensive method.

- Saves time and manpower by avoiding the need to rekey data.
- Eliminates the errors introduced by rekeying.
- Data arrives much faster than it could be by mail, and there is an automatic acknowledgment.
- Imposes a fairly strict discipline on its users (As an example, it may force suppliers to quote part numbers or serial numbers or to supply certificates of delivery where these are required). This may be a disadvantage to some people.

## 12.23 LIMITATIONS OF EDI

Though there are accepted benefits, EDI do not find widespread acceptance because of the following:

- (i) High costs – Applications are costing to develop and operate. Specially new entrants find this more difficult to have EDI.
- (ii) Limited accessibility – It does not allow consumers to communicate or transact with vendors in an easy way. A subscriber must subscribe to an online service called VAN (value added network).
- (iii) Rigid requirements – Needs highly structured protocols, previously established arrangement, and unique proprietary bilateral information exchanges.
- (iv) EDI applications automate any certain portion of the transactions.
- (v) Applications are narrow in scope.

## 12.24 CONCEPT OF EDI

EDI is Electronic Data Interchange which is a very simple concept. It uses direct links between computers, even between computers on different sites, to transmit data to eliminate data sent in printed form. EDI is generally thought of as replacing standardised documents such as order forms, delivery notes and invoices. The technique is highly flexible. EDI's application is vast and used in wide range of industries.

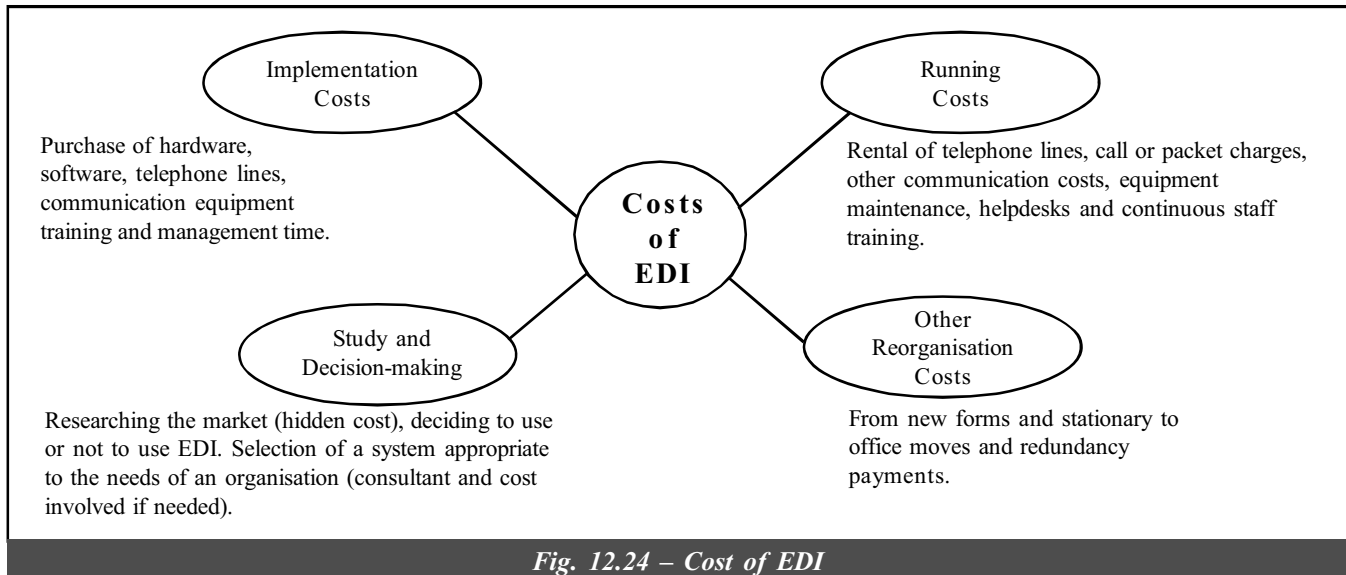
## 12.25 DISADVANTAGES OF EDI

- Since EDI is a structured way of working, companies usually change operating procedures.
- Responsibilities may have to be changed during the introduction of EDI system. Unless this system and the links with other systems are managed well, it is not possible for the data processing department to become involved in production and purchasing decisions.
- Less transparent than paper-based systems.
- Certain EDI systems are highly flexible; others are very simple to implement.
- Users have developed systems to take advantage of the FAX machine which may avoid portal delays. Acknowledgment could be received through FAX.

- How secure is the operation is one of the questions. Are there standards for the type of operation we envisage? What happens when telephone lines or computers fail? How can the backup system work to replace?

## 12.26 COST OF EDI

The cost of EDI could be for the following:



*Fig. 12.24 – Cost of EDI*

There are many soft benefits too, arising from most efficient operation, direct link with customers, and better management controls. These cannot be called as savings, but a decision could be arrived from this to decide whether to adopt EDI or not.

While implementing an EDI system, one has to:

- Review all parts of the operation concerned, and all paperwork and reports generated.
- Decide what actually controls the operation.
- Get rid of any unnecessary operations, reports and paperwork.

EDI allows existing staff to handle a greater throughput.

## 12.27 EDI MODEL

This involves trading partners who want to exchange data from the organisations (may be customer and supplier). There may be two companies with a common customer or two banks whose customers want to deal with one another or between two divisions of the same company, although if they have an integrated computer system, normally this would not be treated as EDI.

Trading partners will have the flow of data between them through exchanges. The simplest and the most common form of exchange is where one partner wants to send a single message to the other and to know whether or not the message received by the other. Further, there may be sequential messages following the first one, in some way linked to the first which would be part of the same exchange, which is similar to the telephone conversation carried on by telephone answering machine between two people who are not present when the other calls. The exchange is divided into messages on which most EDI standards concentrate. The messages, if passed successfully and reliably from one partner to the other, it is said EDI is operated.

Any message has an originator who is the creator of the message and a destination which is the other trading partner. There may be one message to reach several destinations, though the protocol of the EDI system usually not permit this.



## Trading Partners

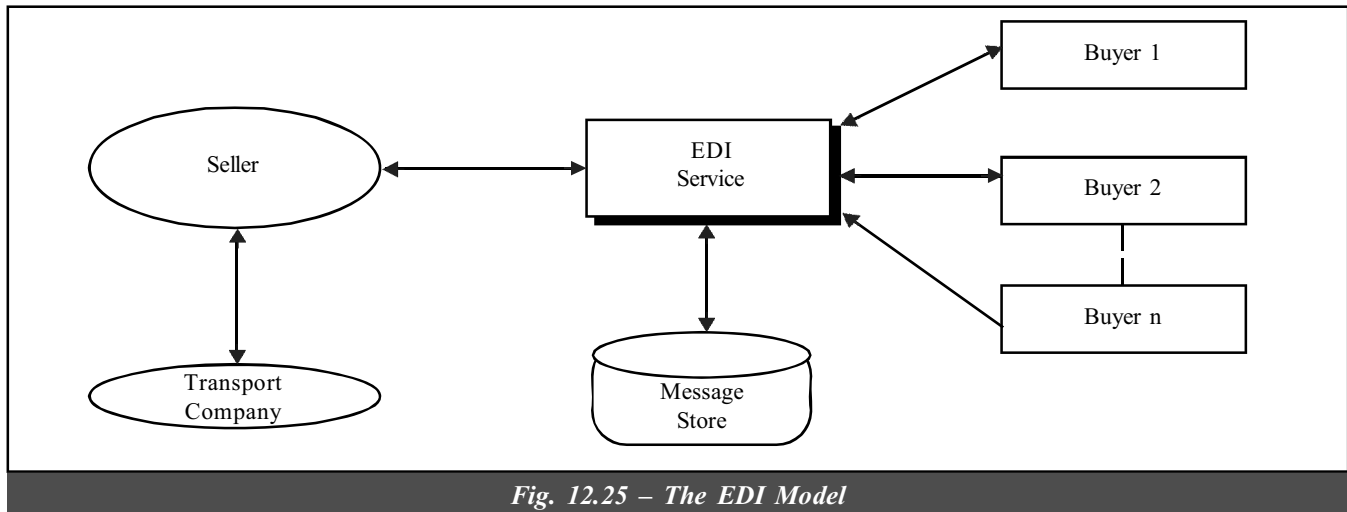


Fig. 12.25 – The EDI Model

**12.28 PROTOCOL**

The protocol which is the part of an EDI standard is the set of rules which lays down how messages are structured, what data is present and the order in which it is transmitted and what additional information is to be provided if required. Further, it specifies limits (character set and the encryption methods allowed). The electrical interface design such as number of wires, type of signals, the type of plug used and the identification numbers and passwords for originators and destinations are allocated. Most known EDI standards for commercial use are GTDI/EDI FACT set, and in USA, the equivalent set is ANSI X.12.

To encrypt information and to prevent unauthorised users to access the same, computers divide the information into blocks and perform a mathematical manipulation on each block, with a sequence of data called as encryption key. To get back the data from the encrypted stage, a further manipulation uses the data encryption key and reverse the process which makes the data to read again, known as decryption with the help of decryption key. Hence, key management is an essential part of any encryption-decryption process. This has already been explained in chapters 'Internet Security' and also covered in 'e-Security' (Chapter 22).

**12.29 DATA STANDARDS USED IN EDI**

The two data standards commonly used in EDI systems are:

- Data Encryption Standard (DES)
- Rivert-Shamier-Adelmann (RSA)

DES standard was developed by IBM for the US Department of Defence and was later on published as a standard. The same key is used both for encryption and decryption of the messages. RSA standard was developed by a group of mathematicians who believed that it would not be possible to devise a code that could be deciphered using a public key without giving away the encryption key.

**THE LEAST YOU NEED TO KNOW**

- ⇒ Define EDI. How EDI benefits business relationships?
- ⇒ An understanding of network-enabled business practices?
- ⇒ What are the advantages, disadvantages, applications, limitations, concept, cost and savings?
- ⇒ Which are the data standards used in EDI?
- ⇒ An understanding of E-commerce development trends in 2018?

**WRAP UP**

- Computer-to-computer exchange of business information has become an increasingly popular form of e-commerce. EDI is considered important because the firms could exchange business information much faster than earlier at least expense and with accuracy than is possible using paper-based systems.
- EDI is used in various organisational functions and in industries like manufacturing, shipping, warehousing utilities, pharmaceuticals, construction, petroleum, metals, banking, insurance, retailing, government, etc. EDI consists of standardised electronic message formats for business documents such as requests for quotations, purchase orders, purchase change orders, bills of lading, receiving advices and invoices.
- In order to switch over to EDI, a company must have computerised accounting records and establish trading partners who agree to exchange EDI transactions. EDI configuration will involve transaction software at all participating sites, with a standard communications package to shuttle the converted documents between locations.
- EDI and e-commerce solutions are designed to enable companies to communicate business data electronically irrespective of the size of the companies with any trading partner, any industry and any application.

**TAKING STOCK**

1. Define EDI. How EDI would benefit business relationships between organisations?
2. What are network-enabled business practices? Explain each of them. Represent them in a sketch.
3. Mention the four key requirements of EDI. How does EDI differs from electronic mail?
4. 'EDI has replaced paper for certain transactions.' Substantiate this statement.
5. What are the applications of EDI? Mention the advantages and disadvantages of EDI.
6. What are the different costs involved in EDI and what savings could be made from EDI?
7. What is the protocol of EDI standard and what purpose it serves?
8. Explain e-commerce as applicable to customer relationship management (CRM).
9. What is your opinion of mobile CRM in the age of Big Data?
10. Explain Web EDI benefits and features.
11. Write a note on E-commerce Development trends in the recent years (2018 and 2019).
12. Write a note on:
  - EDI and E-commerce solutions
  - Cloud-based EDI



# **C H A P T E R 12c**

## ***Wireless Application Protocol (WAP)***

- 12.30 Introduction*
- 12.31 Definition of WAP*
- 12.32 Handheld Devices*
- 12.33 The Future is WAP*
- 12.34 Mobility and Commerce*
- 12.35 Technology Update*
- 12.36 Security of WAP Transactions*
- 12.37 Mobile Company in Four Dimensions*
- 12.38 Wireless Web*
- 12.39 Web Security*
- 12.40 Public Key Infrastructure (PKI)*
- 12.41 Checklist for Security Policy*
- 12.42 Alias Card*
- 12.43 Positional Services*
- 12.44 Wireless Internet as the Way of the Future*
- 12.45 The Wireless Millennium*
- 12.46 The WAP Environment*
- 12.47 WAP in India*
- 12.48 Mobile Computing*
- 12.49 Stages of Mobile Computing*
- 12.50 Third Generation Mobile*

## 12.30 INTRODUCTION

The Wireless Application Protocol (WAP) is a protocol that supports mobile computing. WAP is a standardised way that a mobile phone talks to a server installed in the mobile phone network to define a standard of how content from the Internet is filtered for mobile communication.

It puts intelligence in the WAP Gateway (networks) which adding just a micro-browser to the mobile phones makes them WAP-enabled. WAP's approach philosophy is to utilise as few resources as possible on the handheld device and compensating for the constraints of the device by enriching the functionality of the network. The WAP Gateway translates WML to HTML and *vice versa*.

WAP-enabled devices would help WAP to bring the convergence of voice and data. Content is made available on cellphone tiny screens. It is not compatible with much of the technology that underlies the current web.

To summarise, WAP creates a parallel Internet and limits the content that the end-users are able to get. It is estimated by the International Corporation that approximately 1.3 billion people worldwide will be plugged to web-enabled phones by the year 2000. Hence, the security issues will pose a problem with mobile devices and it is very much desired that suitable procedures similar to the use of firewalls within this gateway are installed. Corporates are therefore opening themselves up to security breaches.

Security issues in mobile computing include:

- Authentication
- Identification
- Secure messaging
- Certification
- Resource control
- Non-repudiation
- Trusted third parties
- Denial-of-service attacks.

Security at the host as well as at the agent level is warranted.

- Disclosure/eavesdropping.
- Alteration/corruption of data or information.
- Copy or replay of content by an intruder.
- Denial-of-service attacks.
- Excessive resource consumption.
- Repudiation problems over completed transactions.
- Spoofing/masquerading.
- Incoming agent tries to gain access to information, corrupts or terminates the agent platform, consumes resources or alternatively devices resources.
- Receiving agent's platform, tries to exact information, corrupt or modify agent, deny service or terminate connection.
- Other agents try to pacify transactions, eavesdrop upon or interfere with an agent's activities.
- Other entities try to intercept information, interfere or subvert the agent's framework.

Agents should be used for:

- Maintaining security policy throughout the enterprise.
- Managing configuration of networked resources.
- Test for operational vulnerabilities.
- To complement PKI search for components of certificate chain during signature verification.
- To monitor resources and collaborate to share enterprise-wide information.
- To take measures in conjunction with other agents.
- Installing anti-virus software to prevent virus writers from distributing malicious code.

- (i) Intel-protected Access Architecture is a standard authentication platform that gives security to mobile devices. This architecture eliminates the need for recreating a driver every time a vendor wants to change or add another level of security to a device.
- (ii) Notebook security takes the form of guarding access to hard disk data with PINs, smart cards and biometric devices (fingerprints reader, etc.), voice IDs and even photo ID for access.
- (iii) e-signs consortium is a cooperative effort amongst software vendors, smart card companies, certificate authority centres, mobile phone equipment companies and network operations. Sign is a mobile business enabler, since it has a SIM (Subscriber Identity Module) card that can be turned into a card reader, e.g., a customer doing mobile banking enters a PIN code into the phone that activates the digital signature capability stored in the phone's SIM card. The phone user has to wait for the bank's verification and for carrying out the transaction. A customer could at the same time be hooked up to the bank call centre. The information would be exchanged via short message service (SMS). The technology will also be made to conform to the WAP (Wireless Application Protocol).

### 12.31 DEFINITION OF WAP

*The Wireless Application Protocol (WAP) is a set of specifications, developed by the WAP Forum that lets developers using Wireless Markup Language (WML) built networked applications designed for handheld wireless devices. WAP was designed to work within the constraints of these devices; a limited memory and CPU size; small, monochrome screens; low bandwidth and erratic connections. WAP is a *defacto* standard with support from more than 200 vendors.*

#### Wireless Application Protocol

WAP is an open, global communication protocol and application environment that empowers users of WIDs to easily access and interact with information and services instantly. WAP provides service interoperability even between different device families. It can be built on any operating system and is designed to work seamlessly with most wireless networks. It is ideally suited for wireless internet and optimises the available bandwidth unlike Internet standards like Internet Protocol (IP). In 1997, Ericsson, Motorola, Nokia and Phone.com formed the WAP Forum – the trade group advocating WAP technology. The goal of the Forum was to bring Internet content and other advanced data services to wireless phones on all platforms. The need was felt for developing an open ended standard which, once created, would be supported by all WID manufacturers. The latest version of WAP i.e., WAP v1.2, supports features like push (proactive sending of information to mobile devices) architecture, wireless telephony, etc. WAP v1.3, soon to be ratified by the WAP Forum, is expected to address the important issue of security. The WAP, specification revolves around Wireless Markup Language (WML), which makes optimum use of small screens of WIDs, allows easy navigation with one hand without a conventional keyboard, and has in-built scalability from two-line text displays through to the full graphic screen on WIDs.

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| Application layer | Wireless Application Environment (WAE)              |
| Session layer     | Wireless Session Protocol (WSP)                     |
| Transaction layer | Wireless Transactional Protocol                     |
| Security layer    | Wireless Transport Layer Security (WTLS)            |
| Transport layer   | Wireless Datagram Layer (WDP)                       |
| Network layer     | Wireless Bearers (SMS, USSD, CSD, CDMA, iDEN, CDPD) |

*Table 12.6 – The WAP Protocol Stack*

### 12.32 HANDHELD DEVICES

Handheld devices are more limited than desktop computers in several important ways. Their screens are small – perhaps a few inches square or able to display only a few lines of text – and they are often monochrome instead of colour. Their input capabilities are limited to a few buttons or numbers or entering data takes extra time, as happens with a personal digital assistant's (PDA) handwriting-recognition capabilities. They have less processing power and memory to work with and their wireless network connections have less bandwidth and are slower than those of computer hardwired to fast LANs.

The Wireless Application Protocol (WAP) was designed to make it easier to create networked applications for handheld devices despite those drawbacks. WAP is a standardisation effort by the Wireless Application Protocol Forum Ltd., an industry association comprising more than 200 vendors of wireless devices, services and tools. The goals of the WAP Forum

is to provide a set of specifications that allow developers to write Internet-enabled applications that run on small form-factor, wireless devices. Typically, these devices are smart phones, pagers and PADs.

### The Problems with Handhelds

A handheld's constraints mean that it is usually impossible to directly port a desktop application to a wireless handheld device. For the same reasons, it is difficult to directly access most websites with a handheld device. Web applications are traditionally designed based on the assumption that visitors will have a desktop computer with a large screen and a mouse. A smartphone cannot display a large colour graphic and does not have point-and-click navigation capabilities. Programmers need to rewrite applications, taking into account the limitations of these devices, and design websites so that handheld users can access them.

But the handheld device market consists of many different devices running on competing operating systems: 3Com, Corps, Palm OS, Psion Plc's EPOC Operating System, Microsoft Corp's Windows CE, Motorola Corp's, FlexOS, Microware Systems Corp's OS-9 and Sun Microsystems Inc.'s Java, for example. Handheld applications also need to run over a variety of wireless network architectures such as Cellular Digital Packet Data (CDPD), Code Division Multiple Access (CDMA), Global Systems for Mobile Communications (GSM), Personal Digital Cellular (PDC), Personal Handyphone System (PHS), Time division multiple access (TDMA), Flex (Motorola's one-way paging protocol), ReFlex (Motorola's two-way paging protocol), Integrated Digital Enhanced Network (iDEN), Tetra, Digital Enhanced Cordless Telecommunications (DECT), DataTAC (an advanced radio data information service network) and the Mobitex RAM mobile data network. In order to create a common programming environment that would let a developer write an application that runs on multiple devices and networks, the WAP Specification Suite was born.

Think of it as the IBM PC of the mobile world, says Carl Zetie, an analyst at Giga Information Group Inc. in Santa Clara, Calif. "It's a single spec that everyone can write to. Instead of having tiny little islands of mobile informations, an user with any handset can access information."

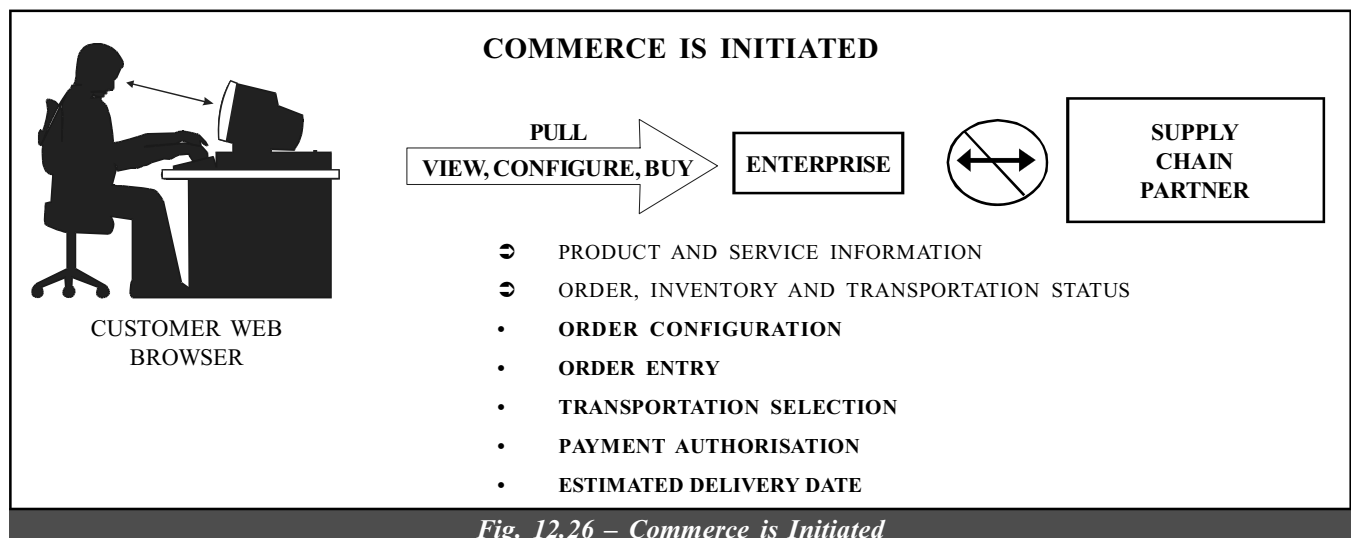
The WAP Forum isn't a standard body, but it does work with international standards organisations and offers its specifications for official recognition.

What makes WAP work as a *defacto* standard is that the major players in the wireless market all support the specification.

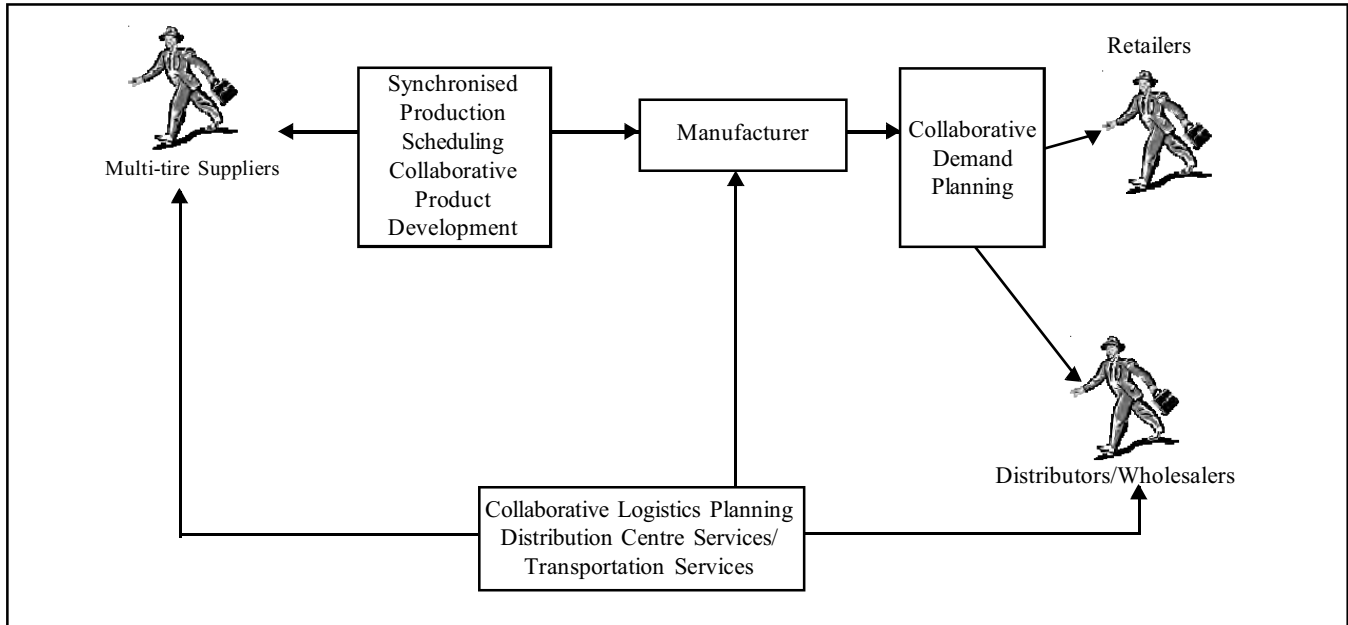
Jill House, an analyst at Framingham, Mass-base International Data Crop. (IDC), lists three of WAP's strong points: "It's got industrywide support, it's non-proprietary and it's well suited to the devices it's being ported to."

WAP is important because more and more information is going out over the wireless network. Recent IDC reports predict that sales of smartphones – just one type of device that supports WAP – will reach 2.6 million units in the US and 539 million units worldwide in 2003.

The WAP Forum has a three stage, public-comment process for including wireless standards specifications in its WAP Specification Suite, now at Version 1.1.



Concepts such as collaborative planning, forecasting and replenishment (CPFR) have allowed companies to redefine traditional roles. Many common administrative tasks, including forecasting, replenishment, filling orders and checking inventory are done automatically with CPFR systems. Sales representatives can now concentrate on such important concerns as finding new customers and enhancing service.



*Fig. 12.27 – Major Supply Chain Collaboration Opportunities*

Following are the three types of functionality to support a collaborative relationship:

- Business framework support
- Process monitoring
- Process management.

The first one represents the relationships set up between business partners, i.e., production schedules, promotional forecasts and work flow.

Collaborative efforts are most likely to be addressed by any or all of these future buyer-seller relationships. Electronic collaboration will differ substantially by a trading partner's role within a supply chain, depending on whether it is a manufacturer, distributor, retailer or third party logistics provider.

CPFR is a more forward looking, proactive standard. It can help eliminate stockouts and excess inventory because partners can respond to demand fluctuations. People are building massive websites with the ability to take orders, and capture data about customers, but precious little attention is being paid to the back-end fulfilment piece. Companies not preparing for this will fail. If their supply chain fulfilment systems are still archaic, they can't handle the new business paradigm.

#### Major Developments in Portable Computing Technology

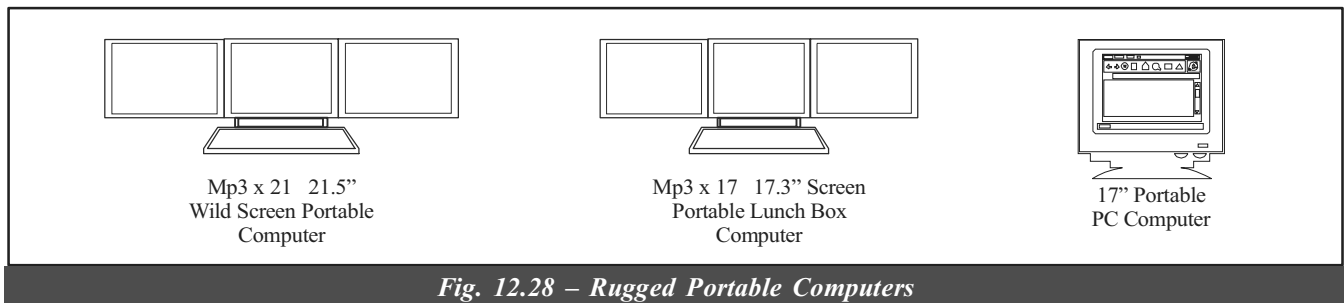
|      |                                                          |
|------|----------------------------------------------------------|
| 1989 | Compaq introduces first notebook PC                      |
| 1993 | Colour displays replace monochrome displays              |
| 1995 | Intel Pentium processors replace 486                     |
| 1996 | The Internet becomes an important business tool          |
| 1997 | IBM introduces a 56-byte disk drive for portables        |
| 1998 | 14.1 in colour displays set up true desktop replacements |
| 1999 | Intel introduces Pentium II CPU in portables             |

### Expected Accelerators to Portable Computing Adoption

|      |                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2000 | Delivery of Intel's Geyserville dual mode processor                                                                               |
| 2000 | Bluetooth-enabled devices come to market                                                                                          |
| 2000 | More USB peripherals come to market                                                                                               |
| 2000 | Universal docking strategies come to market                                                                                       |
| 2000 | 25 Gbytes common in a notebook PC                                                                                                 |
| 2001 | Bluetooth becomes standard in handhelds and notebooks, demonstrations of broadband wireless; Ethernet built-in for most notebooks |
| 2002 | Broad band (>350 bit/sec) becomes widely available                                                                                |
| 2002 | DUD-RW enters the market                                                                                                          |

### 2002-2018

In the recent days, we have 'rugged portable computers, which provide support in areas where it should be impractical to haul a computer, separate monitor, keyboard and a mouse. Portable rugged computer (also called as lunch box computer, portable workstation, or luggable computer) is designed to do just that. These rugged portable computer offer the same functionality of a desktop or portable computer, with the advantages of all necessary peripherals, multiple hard drives, card sets and displays built in.



*Fig. 12.28 – Rugged Portable Computers*

### Others:

- Computers with waterproofing casing (HP Joranda Portable PC)
- Removable and reversible display device for portable computer
- Information communication technology revolutionising tourism
- Portable electronic product with a bracket
- Portable computer with separable screen
- High mechanism for cover panel of portable computer including slide mechanism
- Portable computing device having an adjustable hinge
- Compact portable computer having a risen that forms when a cover is opened
- Computer with several portable computing devices such as handheld, screens (varied), method for managing (howling), docking station, PCD battery, adjustable hinge, LCD screens, etc. (LCD is Liquid Crystal Displays). LCD is the technology used for displays in notebook and other computers.

Computers using social technology – an area of computer science that is concerned with the intersection of social behaviour and computational systems. It is based on creating or recreating social conventions and social counters through the use of software and technology.

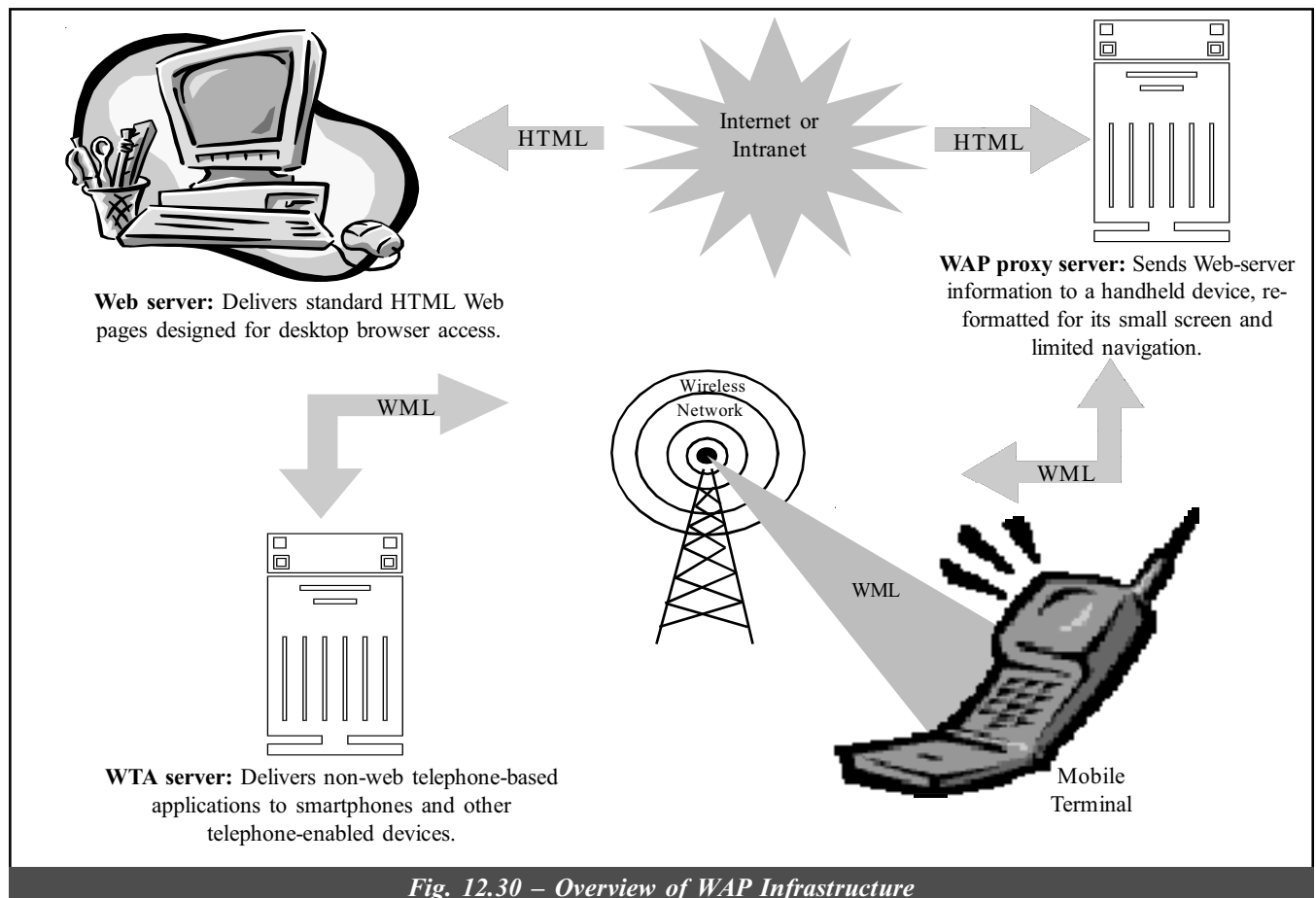
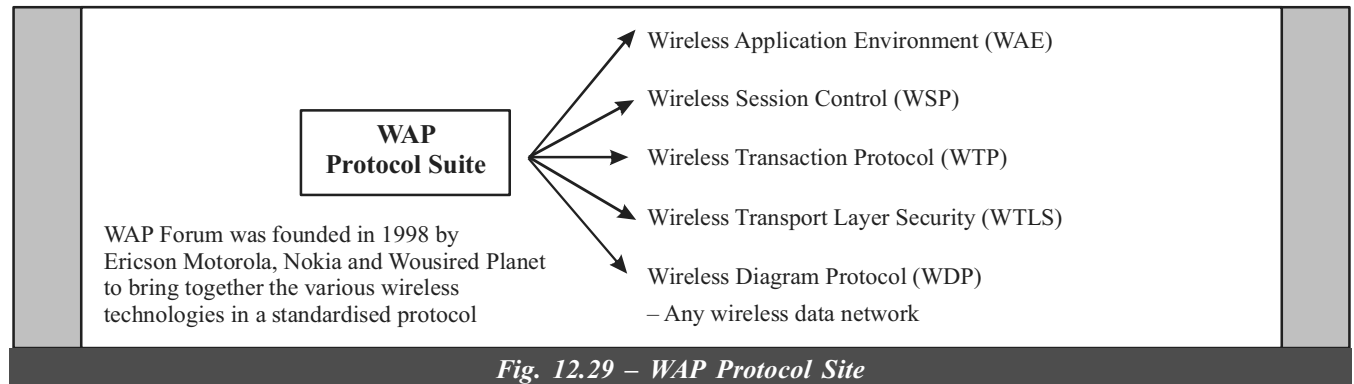
### 12.33 THE FUTURE IS WAP

WAP or Wireless Application Protocol is the gateway to a new world of mobile data. It provides a universal standard, which enables users to easily access web-based interactive information services and applications from the screens of their mobile phones. These include both consumer and corporate solutions.



Wireless Application Protocol (WAP) provides a universal open standard for bringing Internet content and advanced Value Added Services to mobile phones and other wireless devices. As such, WAP is the beginning of the Total Connectivity Age.

WAP achieved some popularity in the early 2000s. In 2010, it is superseded by more modern standards. A reengineered 2.0 version was released in 2002 (WA 2.0).



### 12.34 MOBILITY AND COMMERCE

Most of us think WAP stands for Wireless Application Protocol. But a good number of wireless technologists themselves prefer calling it 'Wasteful Application Protocol!' It is 'finding solutions for problems that don't exist,' according to one analyst who thinks no one in the world is that desperate to buy a T-shirt through his cellular phone!

Before the introduction of WAP, mobile service providers had limited opportunities to offer interactive and services, but needed interactively to support internet and web applications such as e-mail, stock prices, news and sports headlines. The Japanese i-Mode system offered another major computing wireless data protocol.

In the cellular phone's display screen, one can see information only if they don't exceed 160 characters. And typing the URL and getting connected to the Net takes a very long time. In fact, over 75% of the users of i-Mode, a leading Japanese WAP service provider, have said they never use their instruments for browsing. Internet will never be desirable on the phone, or for that matter on any personal digital assistant (PDA).

On the other hand, jumping into the bandwagon are the new m-commerce (mobile commerce) players, who are encouraged by predictions that wireless handheld devices would account for over 40% of online time in another five years. The population of wireless gadgets will explode to touch nearly 500 million around the same time. The services offered by phone companies that are based on technologies like short message services (SMS), global system for messages (GMS), WAP... have been getting good reception especially in Asia and Europe.

But isn't it novelty and not utility that drives initial reception of any new technology?

Consumers are told that through the WAP-enabled Ericsson or Nokia, they can get news updates, horoscope, stock quotes, etc. This is what a pager does today; the only difference being that WAP phone is interactive and not one-way, meaning, one can browse the Net and reply to e-mail messages.

Business is told that WAP would be a boost to the retailing industry and venture capitalists are willing to listen to any new m-commerce centric business plan of a bookshop or a toy store.

Consumers are not sure why they should buy a phone that acts as a PC and a browser instead of buying a PC, which already acts as a phone with a browser. The usage pattern of WAP phone owners – only 20% use cell for visiting home pages – really discourages the retail segment.

The sceptics of m-commerce have more logical reasons for being sceptics. One main reason is that E-commerce is yet to be proved successful when it comes to B2C and m-commerce undoubtedly banks heavily rely on this B2C segment.

On the positive side, it is true that the capabilities of the cellular phone could and should be enhanced. 'We want people to get used to using their phones for more than just making calls,' say WAP companies. Certainly they will. At best, m-commerce could complement e-commerce. But what really matters is the convergence calculation: to what extent can a cellular can be used as a browser?

## 12.35 TECHNOLOGY UPDATE

### Wire-free World (Bluetooth)

Soon electronic office devices will communicate with each other over short distances without wires. Bluetooth, the new wireless communications technology, will eliminate cable chaos and pave the way for potential mobile business.

More than a thousand years after his death in AD 986, the Danish king Harald Bluetooth is the talk of the town. That's because the belligerent Viking's name has been adopted for a new communications technology that is grabbing attention worldwide. The name was chosen by the members of the United States-based Bluetooth Special Interest Group established in Santa Clara, California in 1998. Their reasoning: they found the wireless interconnection of telephones, fax machines, printers, computers and consumer electronics to be as challenging as unifying the quarreling kingdoms of Norway and Denmark in King Harald's time. All the devices containing a Bluetooth chip can communicate with each other and transfer data without wires. Within a 10 meters range, they can automatically establish and maintain contact via the radio signals. Because the waves can pass through solid objects, networks of this type, called "piconets" even function when, for example, organisers are tucked away in briefcases. But developers have not limited their imagination to merely the office. They have also set their sights on 'hotspots' – the areas where business travellers tend to congregate. In the future, business travellers will have access to any desired network – be it the Internet, intranets, local area networks (LANs) or wide area networks (WANs) – at airports, train stations, hotels and other similar places. And all of this will be possible regardless of the kind of terminal they use. It is also conceivable that the system could be used in sales showrooms so as to provide customers with information on availability of products, delivery times and methods of payment. No longer would a customer need to access a computer network. Instead, customer could call up the information needed on the wireless palmtop, simply by connecting it to the company's LAN via the new 'Bluetooth'.

*Bluetooth is a computing, networking and telecommunications industry specification that enables mobile phones, computers and other WIDs to 'talk' to each other using radiowaves.* The Bluetooth special interest group, which includes

Nokia, Ericsson, IBM, Intel, 3Com, Lucent, Microsoft, Motorola and Toshiba, is driving the development of this technology and bringing it to the market. The technology requires the inclusion of a low-cost (\$5) transceiver chip in each device. According to the Gartner Group, 75% of mobile phones around the world would be equipped with the \$5 Bluetooth chip by the year 2004. Products with Bluetooth technology are expected to take off in a big way in India early next year. The applications of Bluetooth are numerous. Data and notes can be transferred from Bluetooth fitted PDA to the PC, and downloaded MP3 files can be transferred from the PC to the MP3 Player without physically connecting them. It also acts as a payment tool by debiting the user's bank account as soon as a transaction is completed.

### 12.36 SECURITY OF WAP TRANSACTIONS

A number of concerns are being raised about the security of WAP transactions. This is being blown up beyond reasonable proportion. Certicom, one of the world's leading security solutions providers, is now offering products that ensure that the security of such transactions are spacious. This is as secure or insecure as shopping on the Net, which is as secure as insecure as giving the waiter your credit card to swipe at a restaurant. Nothing can stop him from noting down the details of making extra copies of the charge slip. So how is the current process any better or safer?

### 12.37 MOBILE COMPUTING IN FOUR DIMENSIONS

Mobile computing has already been explained in Chapter 4 (paras 4.1 to 4.5).

Companies should assess their remote access needs and plan the architecture from the start of the activity. Many mobile phones can be configured by remote control.

WAP services can be switched on by sending the phone a series of commands which are then recorded on its internal SIM card. Security measures are built into WAP in such a way that only network operators and the makers of SIM cards could reconfigure it. WAP Forum gives specialisation for a standard method for small limited resource devices such as cellular phones to access the Internet. In between the WAP phone and the Internet is a WAP proxy which provides a gateway between the wireless and the Internet environments. The WAP proxy tokenises the data before transmitting it to the phone, thus achieving a measure of data compression.

The expansion of mobile computing is in four dimensions:

- (i) Wireless delivery technology and switching methods
- (ii) Mobile information access devices
- (iii) Mobile data internet working standards and equipment and
- (iv) Mobile computing based business applications.

These are explained below.

### 12.38 WIRELESS WEB

The wireless internet allows full access to the web via a mobile handset, or any mobile device such as a personal organiser. While WAP is a specific protocol which allows one to surf by using WML as the scripting language, the wireless internet is not restricted to any particular protocol or hypertext markup language. Now, compact HTML is a much closer language to HTML. If you look at rediff.com, cnn.com and yahoo.com, you are looking mainly at HTML. That could be shown as E-HTML much more easily than via WAP.

| Wireless Delivery Technology and Switching Methods                                                       | Mobile Information Access Device                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cellular, radio, paging and satellite LAN communications information to mobile users any where any time. | Explosion in the variety of access and wireless equipment – laptops, notebooks and other promise to furnish portable computers with large memories and powerful processors to handhelds or palmtops – personal digital assistants or personnel communications. |

| Mobile Data Internet Working Standards and Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mobile Computing Based Business Applications                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Voice-oriented cellular systems were not designed with data transmission in mind. Weak connections, frequent noise and momentary dropouts that cellular phone users move from one cell to another or encounter physical obstructions such as bridges and tunnels are an inconvenience for voice calls. These can easily corrupt data calls. Another problem is insufficient capacity in terms of the number of available frequencies. To solve these problems, work is done in the area of digital cellular standards that support data communications over cellular lines.</p> | <p>Two types of applications frequently used are: (i) mail enabled applications and (ii) information or data services to mobile users. Mobility and portability are creating an entire new class of applications, new markets, combining personal computing and consumer electronics. Vertical niche applications available are:</p> <ul style="list-style-type: none"> <li>• Vehicle dispatch and routing and</li> <li>• Inventory and package tracking.</li> </ul> |

*Table 12.7 – Expansion of Mobile Computing in Four Dimensions*

Any and every security system developed by the human mind can be broken by another human mind. No system is unbreakable given the time and incentive to do it.

### 12.39 WEB SECURITY

Majority of the web servers combine the strength of security socket layer or SSL-complaint encryption and authentication technology with the advantages of the security model. Applications can be isolated into pre-defined secure areas, which significantly reduce the enterprise's exposure to security risk. SSL is the standard for secure communications on the Web.

### 12.40 PUBLIC KEY INFRASTRUCTURE (PKI)

The bundle of technologies developed under the control of Internet Engineering Task Force (IETF) is known as Public Key Infrastructure (PKI). PKI can be used within the organisation to establish and enforce security policies. It can also be applied across the extended enterprise through extranets or out onto the Internet. PKI has a broader scope than security technology, because it addresses the two key issues: Identification and Encryption. It uses tamper-proof digital certificates. These certificates are issued by trusted third parties and are easily stored on computers, mobile phones or smart cards. These certificates enable individuals to log on to security systems or digitally sign documents. They also contain public key information and enable parties to encrypt high value transactions on the Net.

Opening up IT systems to third parties brings productivity benefits. The security issues are also raised not just in terms of PKI and digital certification, but in terms of multiple level access. It is necessary to implement a strongly trusted network infrastructure and to ensure availability 24 hours a day, 365 days a year.

### 12.41 CHECKLIST FOR SECURITY POLICY

The security policy adopted as part of an e-commerce programme should be focussed on the following checklist:

- Application
- Strong authentication
- Strong encryption
- Network
- Operating system and
- Database.

Microwave ovens will be web-enabled so that while you are heating your dinner in the microwave, you can ask the Internet to order your dessert. (At the moment, m-commerce and e-commerce using mobile devices is rather limited). Technology is not the only reason why M-commerce has not taken off. A few security issues also remain as a major stumbling block.

When you are making a purchase on the Internet today, do not worry about the security side of it. There are certainly sites out there which are scanning customers and taking credit cards. But users do not know who are the bad ones. So, they get scared and are hesitant about internet commerce in general.

### 12.42 ALIAS CARD

Although a significant number of people are into transacting over the internet without worrying about using credit cards and if you want everybody to be comfortable with this issue, then obviously something must be done to safeguard against misuse and malpractice. The use of one-time cards would be better. Each user is given a number of card aliases, each having a distinctive number and mapped onto the cardholder's single PIN number. On each use of the alias card over the Internet, it is permanently discarded never to be used again by that user. This guarantees the safety that users are seeking, while maintaining flexibility of operations. This would allow transactions on the Internet as well as WAP, landline and wireless e-commerce.

### 12.43 POSITIONAL SERVICES

Wireless offers one more feature that landline does not. It offers what are called positional services. Ideally, you could surf into any local site that you know, say rediff.com and find out the name and location of a particular type of restaurant. The directions that you would receive would be relative to your positional coordinates. You have a set of GPS coordinates associated with your mobile phone. So, you send off your coordinates and the site is clever enough to say which one of all the restaurants within a reasonable distance (say half a KM radius) have tables available tonight. Would you like to book one? You press yes, non-smoking for five and you wander along and have your meal.

The other important point in favour of the wireless internet is that you would be able to quickly pinpoint the exact information, any time and any place. With location specific information, you would be able to zoom into your target quickly and go there physically to pick up the item of interest. The main frustration on the net is that it takes longer to buy something and it is hard to target as there is so much on the Net. Here, it is location-specific targeting which means you get things much faster.

### 12.44 WIRELESS INTERNET AS THE WAY OF THE FUTURE

Some people see the wireless internet as the way of the future and is pushing hard to develop a couple of related technologies like Bluetooth, UMTS, 3G GPRS, iMode, EDGE, etc. These will allow hands-free operation of mobile devices and video conferencing over mobile handsets sporting much larger screens. Brief details of the above technologies are as indicated below:

- (i) Bluetooth is a global defacto standard for wireless connectivity. Based on a low cost, short range radio link, Bluetooth cuts the cards that used to tie up digital devices. When two bluetooth-equipped devices come within 10 meters to each other, they establish a connection. Your laptop could send information to a printer in the next room or your microwave could send a message to your mobile phone telling you that your meal is ready. Bluetooth is likely to be the standard in tens of millions of mobile phones, PCs, laptops and a whole range of other electronic devices.
- (ii) UMTS (Universal Mobile Telecommunications System) is the so-called third generation (3G) broadband, packet based transmission of text, digitised voice, video and multimedia at data rates upto and possibly higher than 2 megabits per second (Mbps). It will offer a consistent set of services to mobile computer and phone users, no matter where they are located in the world. Based on the global system for mobile (GSM) communication standard, UMTS, endorsed by major standard bodies and manufacturers, is the planned standard for mobile users around the world by 2002.
- (iii) GPRS (General Packet Radio Services) is a packet-based wireless communication service that promises data transmission rates from 56 Kbps upto 114 Kbps and continuous connection to the Internet for mobile phone and computer users. 3Gs GPRS is third Generation GPRS.
- (iv) i-Mode is a wireless technology developed by the Japanese company NTT DoCoMo that enables users to access internet services via their cellular phones. i-Mode (i stands for information) is also based on packet data transmission technology. This means that i-Mode is always online, and therefore users are charged only for how much information they retrieve, not how many minutes they are using it for. i-Mode can be used to exchange e-mail with computers, personal digital assistants (PDAs) and other i-Mode cellular phones.
- (v) EDGE (Enhanced Data GSM Environment), a faster version of the global system for mobile (GSM communication) wireless service, is designed to deliver data at rates upto 384 Kbps and enable the delivery of multimedia and

other broadband applications to mobile phone and computer users. Edge is expected to be commercially available next year. It is regarded as an evolutionary standard on the way to Universal Mobile Telecommunications Services System.

## 12.45 THE WIRELESS MILLENNIUM

The backbone of the ongoing IT revolution is the Internet. The showcase of this ubiquitous network is the World Wide Web (WWW). Now, after a decade of its presence, one might ask the question, what is beyond the Web? The answer is loud and clear – we are being led towards a Wireless Web. The all-pervading and omnipresent wireless web is in the making.

The wireless web is the consolidation of people's aspiration and craving for information to be available anywhere anytime and the resulting mobility could even be the key to success. PDAs (Personal Digital Assistants) and a host of similar products now have the capability of being connected to the Internet wirelessly. Even more interesting is the marriage of the mobile phone with the Internet. The wireless mobile devices are soon going to be the universal personal interface to information. The growing demand for mobility has spawned a great demand for wireless products and services. It is estimated that by the end of year 2003, the number of handheld wireless devices will be over one billion worldwide and the mobile mode of internet access would surpass the use of PC for internet access.

Now, it is the turn of the humble mobile phone to revolutionise the telecommunications world. If not replacing the traditional Plain Old Telephone Services (POTS), it competes with it and complements it too. It originated as analogue mobile voice telephony and evolved with overlaid circuit switched data service to the present-day packet-switched data service offering simultaneous voice and other value-added services. The marriage of the mobile phone and the internet has changed mobile telephony into mobile computing and this has opened up a plethora of possibilities. Even as it evolves, the mobile telephony strives to match up its capabilities with that of the internet. One of the results of this is the new technology called WAP (Wireless Application Protocol) – much talked about by everyone.

WAP is an application environment and a set of communication protocols for wireless devices designed to enable manufacturer, vendor and technology independent access to the internet and advanced telephony services. The current implementation of WAP over the GSM networks offers a bandwidth of 9.6 Kbps.

Interestingly, WAP is not the only flavour of wireless technology solutions. For example, there is a competing technology called i-Mode promoted by NTT DoCoMo of Japan. In fact, there is a wide array of carrier and technology choices like CDMA, TDMA, CDPD, GSM, GPRS, etc. A hidden cost on the users on account of the multiplicity of emerging carrier technologies would be the handset incompatibility. Multi-system handsets will cost more while cheaper single-system handsets will be tied up to a single service type. However, various carrier technologies themselves are evolving to reach the 3G (third generation) cellular telephony specifications designed to give an even richer user experience.

Apart from the mobile phone, use of other types of handheld wireless devices for mobile web access is also gaining popularity. The Palm Pilot, now called Palm, is the present market leader in the PDA segment. The latest in PDA category offers internet surfing, along with other features including mobile telephony. PDAs that behave like cellular phones and mobile phones that behave like PDAs may not necessarily have the risk of losing their identity because both have been designed for different user experiences.

The universal spread of wireless technologies means extended enterprise to the corporate world, mobility to the masses and the integration of every link in-between. Wireless solutions, as a means to extend the information enterprise, could be varied. Whether verticals or real-time application integration solutions for industries, the possibilities are immense. The power of extending the information enterprise has been proved many times over by wireless solutions for Supply Chain Management, Sales Force Management, Field Force Automation and by other point killer applications brought out by the leaders in the wireless solutions industry.

The concept of Extended Wireless Enterprise surrounds the Extended Wired Enterprise paradigm. Data rate capability offers spanning the 28 Kbps to 11 Mbps range covering a wide range of physical distances and thus makes this extension possible. A wireless gap analysis would reveal the cracks that would develop soon in an organisation if it does not plan to take the Extended Wireless Enterprise route.

Yet another step in wireless connectivity is the Bluetooth technology. It is a protocol for wireless interconnectivity among digital devices and therefore the interconnecting cables could be totally done away with. Connectivity between such devices is secure and has a bandwidth of approximately 720 Kbps. The whole bluetooth technology comes packaged within a micro chip and the chip's cost might soon come down to US\$ 5. Whichever digital equipment that can accommodate this chip can be bluetooth-enabled.

Software applications designed to quench the need for mobility will rule the roost. Streaming multimedia, alert services, location and positioning services, voice recognition, etc. are all waiting to be delivered through the new mobile delivery channels. Voice XML, SyncML and other technologies promise a whole new dimension to the mobile experience. Any product that can be digitised can be delivered literally to your hands through the wireless web. Perhaps, the already thin boundaries among computing, communication and networking will totally fade away and the ultimate convergence of these fields makes the need for wearable computers an imminent necessity of the coming days.

Ultimately, where are these technologies taking us? It is largely left to the market forces of the new internet economy to decide. The present reality is that when your mobile phone is 'dotcom ready,' the world will be wirelessly yours.

The user of a wireless device, such as a mobile phone, sends in a request for information to the Web Server through the WAP Gateway. The requested information is delivered using WML format, which if already is compatible can be downloaded directly. On the other hand, if the information is still in the HTML format, a filter is used to convert the same to WML. However, being WAP-enabled is necessary in order to exchange information.

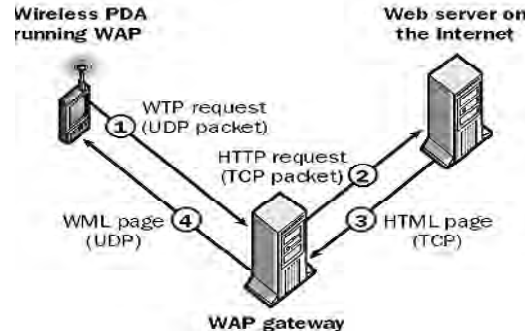


Fig. 12.31 – The WAP Environment

## 12.46 THE WAP ENVIRONMENT

Wireless Application Protocol (WAP) is a technical standard for accessing information over a mobile wireless network. A WAP browser is a web browser for mobile devices such as mobile phones that uses the protocol.

### Mobile Internet

Obviously, services that you use while being “mobile” are in general better suited for the mobile internet than the classic “web”. Some great applications are starting to emerge, but this is just the tip of the iceberg. Some of the key WAP application categories include:

#### *Wireless Access to Internet Content*

Consumers benefit from immediate interactive access to the information they need at that moment, location-specific shops, theatre, map, taxi and GPS-driven locator.

#### *Wireless Access to Corporate IT Systems and Extranets*

Corporations can offer new channels for their services and also create totally new services for their mobile customers in addition to directory services and intranets.

#### *Wireless Access to Personal Information*

Wireless device users can access their e-mail, calendars, ‘to-do’ lists, and even screen text headers for their voicemail messages. Immediate interest areas are like information on sports, news and horoscopes.

#### *Intelligent Telephony Services*

Carriers can offer their customers secure access to their personal and other customer-related information, billing and other databases in the Intelligent Network.

### M-commerce

Access to corporate B2B mission critical legacy applications include inventory, sales automation and SCM applications. Day-to-day transaction such as buying/selling goods and services.

## 12.47 WAP IN INDIA

It's less than a year since mobile operators and Internet companies discovered that the cell phone was a great way of expanding the scope of the Internet. And less than a few months since mobile subscribers around the world actually surfed through WAP-enabled Internet content through a WAP phone. India has already ventured into this exclusive market.

The speed of all this has been phenomenal, thanks to the intense market pitch by both multinational cellular giants and local WAP product vendors. All it took was six months and the market today has undergone a drastic change. Since then, companies like Nokia and Ericsson have kicked off this new ball game. Phone.com was also recently heard of being involved in a major wireless Internet service roll-out in India. On the other hand, you can also see young Indian start-ups like Integra Micro, Taruni Technovision and SAS Institute competing with the hardboiled MNCs. Among systems integrators, e-Capital, Planetasia.com, IBM and CMS stood out as being active on the WAP front.

The market's demand of WAP trained professionals is steadily growing and its potential is enormous. If you are interested in this technology, learn it immediately, before the demand runs out, says the experts.

## 12.48 MOBILE COMPUTING

As the computer age advances, huge spaces occupied by computers like mainframes are moving towards better times; the size of the computers are becoming smaller and smaller; and graduated to a level where nomadic computing has become a reality! E-commerce is now moving towards M-commerce or Mobile commerce which has been enabled by:

- Portals/powerful computers
- Reliable communications and
- Increasing bandwidths.

Mobility is the desired characteristic in today's business as we are travelling on the Information super-highway, overcoming all barriers found during communication and distance and carrying a niche in the global market.

Mobile agents are needed:

- To add a new communication paradigm to traditional network communication mechanisms.
- Because of specific advantages found in remote programming or client-server systems as compared to the classical mechanisms.
- Into the widespread and well accepted web environment for seamless integration, when used in a heterogenous networking environment such as the world wide web.
- To integrate themselves effectively into the web and the use of Java cards to allow a mobile agent to store and transport data security for e-commerce purposes.

## 12.49 STAGES OF MOBILE COMPUTING

- First, it made its way with Analog devices to have only voice calls limited to a specific location.
- The mobiles started facilitating fax, data messaging and voice access over continents with the advent of the digital era.
- The latest multimedia technology involves high speed transfer through mobile devices allowing video, audio and Internet through a mobile terminal.

Mobile phones are extensively used today for:

- Messaging
- Accessing company data and
- Numerous mobile commerce services.

Service providers in view of the above demand take an opportunity to offer value-added services by providing security and confidentiality to the interchange of voice and data.

Mobile commerce offers organisations:

- Leaner structures
- Optimisation over geographical limitations
- Boosts the morale of employees who do not always have to undertake stressful journeys to the place of work and
- Give the employees an option for a remote reporting culture in a mobile environment.

Potential security risk with the future mobile services would be large. We have seen many executives carry and use laptops with a dial-up connection to corporate e-mail or Intranet with little or no password protection or encryption. In such



a situation, security breaches would be there. Users are concerned about the risk of data being intercepted on the line as several data transferred are of sensitive type and disclosure can be even fatal. Theft is also other concern wherein thieves used to steal portables for the value of the hardware and even more for the value of the information they contain.

Mobile phones in the future would have more utility which can:

- Access the Internet
- Empower the employees to receive e-mails, faxes and voice data and
- Keeping the employees informed even at the remotest of locations.

These devices along with several advantages would bring lot of concern for data access and security where IT organisations still have reservations on the possible consequences of stolen devices and sensitive files going into wrong hands.

## 12.50 THIRD GENERATION MOBILE

### (i) Problems and Prospects in India

Fundamental developments in telephony products and services are referred to as “Generations”. The 3G (Third Generation) history can be traced back to 1985, when International Telecommunication Union (ITU) announced its Future Public Land Mobile Telecommunication Systems (FPLMTS) initiative. In the year 1996, FPLMTS was known as International Mobile Telecommunications 2000 (IMT-2000). Here, 2000 refers to both the decade of intended introduction and the newly assigned frequency band of 2 GHz. Its main objective was to provide a broadband, inter-operable, worldwide solutions based on multiple technologies offering technologies with greater frequency. This objective makes mobile internet access, multimedia services and the integration of different mobile networks, including Mobile Satellite Systems (MSS) possible through universal coverage.

### (ii) Third Generation (3G)

3G will make possible most of the personal communications, be it in the form of phone calls, data images or live videos. Thus, it will open new opportunities to network operators for bringing new players into the communication market and offer end-users new services and applications, which will enhance their personal and business lives, irrespective of place or time. The entire world will stand on the verge of an era of unparalleled user control and benefits from 3G technologies. Third generation will see a shift of focus away from technology, and towards applications and services. The enormous potential of 3G will support 2 Mbps for special applications, through the frequency in the range of 140-384 Kbps will be common. Handsets, developed through innovative technologies, will be capable of supporting/handling different services and user applications, and also allow transmission between multiple networks to help realise the dream of – “Anywhere-Anytime”.

### *Third Generation (3G) Networks*

High-speed digital data is the Third Generation (analog was the first generation, and digital, second) and an evolutionary chain of solutions that bring increased speed, reliability, security and availability to mobile services. 3G technology is dependent on bundling services to offer key value to the consumer. By 2002, 3G networks like the UMTS (Universal Mobile Telecom Systems) will gradually ramp up data speeds over WAP-enabled WIDs to 40 times today's norms and comparable to that of a desktop PC. At such speeds, things like simultaneous video, e-mail, and phone conversations are possible. With the Indian mobile telephony networks already planning implementation of 3G GSM (Global System for Mobile Communication) systems, mobile phones can be comfortably used to access Internet and corporate networks. This technology would make WIDs, especially WAP-enabled mobile phones, the right medium for the m-commerce revolution in India. In the future, General Packet Radio Switching (GPRS) dramatically speeds up access to the Internet, which will bring down tariffs as well as operating costs for operators because there is no need to dedicate one circuit for one call. Third Generation Partnership Project's (3GPP) organisational partners have agreed to cooperate in the formulation of technical specifications for a third generation mobile system based on further evolved GSM core networks and radio access technology. This will develop multimedia subsystem domain capabilities to allow the creation and introduction of multimedia services, and have a flexible service development to enable operators to introduce services as they see it from their vantage point of business goals.

### (iii) Main Drivers/Key Applications

Whatever be the service, 3G networks will be able to support it. This unparalleled capability will help mobile operators/ service providers to enhance their business image, reduce churn, attract new subscribers and increase traffic volume per subscriber through cost-effective measures, as they will be able to create innovative services on the top of existing networks

and investments. One can imagine-with 3G/IMT-2000 in place – walking into a home and seeing dozens of computers, PDAs and other devices all connected to the Internet.

Real-time video-telephony, for example, will enable a “virtual presence” where doctors and other experts can give advice on situations from miles away, people will be able to take part in meetings without being present, etc. Location tracking will be linked to information services such that customers will not only be able to get information on cinemas and restaurants, but also, will be able to look at maps showing the position of the restaurant or theatre and even view video clips of particular films.

Thus, the increased bandwidth and 3G technology will enable the customers to use their phones for much more than just making calls. It is clearly evident that there is no “Killer Application”, instead, technology is helping to create a “Killer Environment”, where a large range of services and applications can be developed and offered.

#### (iv) Business Case

The shift 3G technology and related applications will create changes for all those involved in the value chain. The wide range of new applications and services will encourage greater airtime usage from customers, thus increasing operators’ revenues. Customers will be able to take advantage of a huge range of services and will tend to develop personalised service profiles, which will increase loyalty and reduce churn. The future business is – “Anywhere-Anytime” mobile communications – not airtime through a bit-pipe. Cellular operators/service provider, etc. should opt to capitalise on their well-established position in the emerging industry by opening wireless portals that incorporate e-commerce and transaction centres. By doing so, operators can:

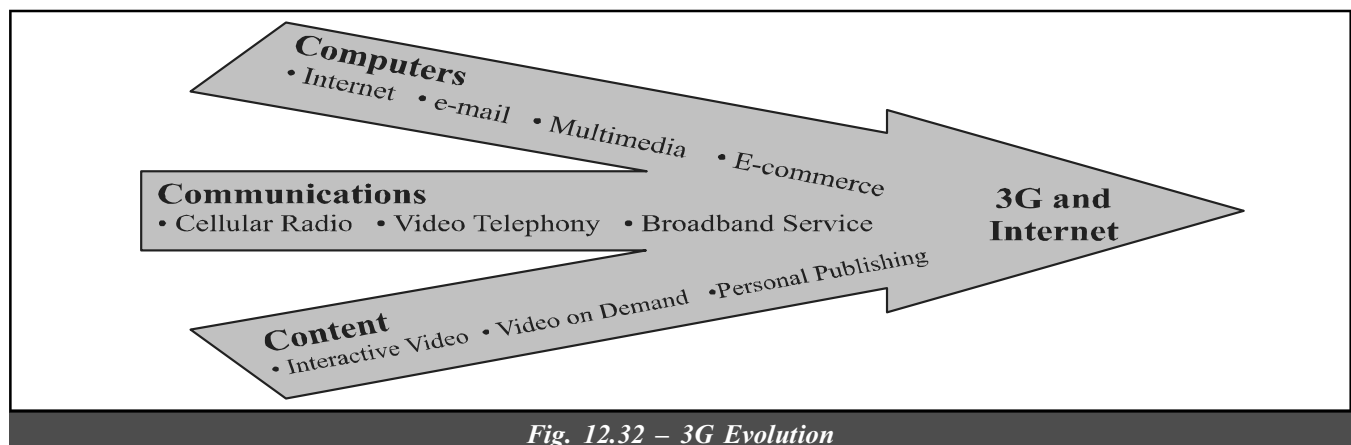
- Differentiate themselves from the competition
- Reduce churn
- Improve their overall business image and
- Enjoy large increases in the datacom traffic.

Business cases revolve around two basic issues – revenue sources and costs incurred.

A typical 3G business case is shown in Figure 12.32. Apart from other things, a 3G model will rest on the super-position that the Internet will dominate.

#### (v) Key Technology Enablers

Open standards based technologies such as HSCD, GPRS, EDGE, WCDMA and IP in general, will be the basic building blocks of global IP mobility. Global IP mobility will provide a ubiquitous network infrastructure providing Internet, corporate Intranet services as well as voice and video. This will enable the dotcoms to move towards mobility, making wireless communications possible, not only over wide area cellular networks, but also LANs and “hot spot” areas through the use of radio access technologies. Use of IPV6 (Internet Protocol Version 6) will create a higher quality and secure next generation Internet.



#### (vi) Be the Leader

Voice and data networks are converging at a blinding speed, changing the way people work, live, play and learn. Therefore, cellular operators/services will ultimately emerge as successful in their area of business. Through the early

introduction of wireless data, operators will gain invaluable experience in marketing and billing for such services and also customer awareness of new services will grow.

### (vii) Learning from the Past

At present, Indian Cellular Operators are completely aware of the challenges and key business opportunities. Some of the key areas that need improvement are:

- *Innovative and cost-effective implementation of the network:* Cellular operators are aware of the strengths/weaknesses of a particular technology, hardware, software, operational support, migration path, etc. as far as 2G systems are concerned. However, they still have to learn and know a lot more about 3G technology, as it is different from 2G in the following respects:
  - IP-based network.
  - Manages and monitors higher layer infrastructure.
  - Manages IP address assignments.
  - Networks are distributed – network elements require more intelligence. End-to-end network management.
  - QoS becomes more complex for 3G.
  - New faults, measurements and configurations.
  - Additional automation required due to increased networks and services/applications complexity.
- *Right selection of vendor:* This is going to be one of the key parameters of success for the right deployment and investment of futuristic technologies.

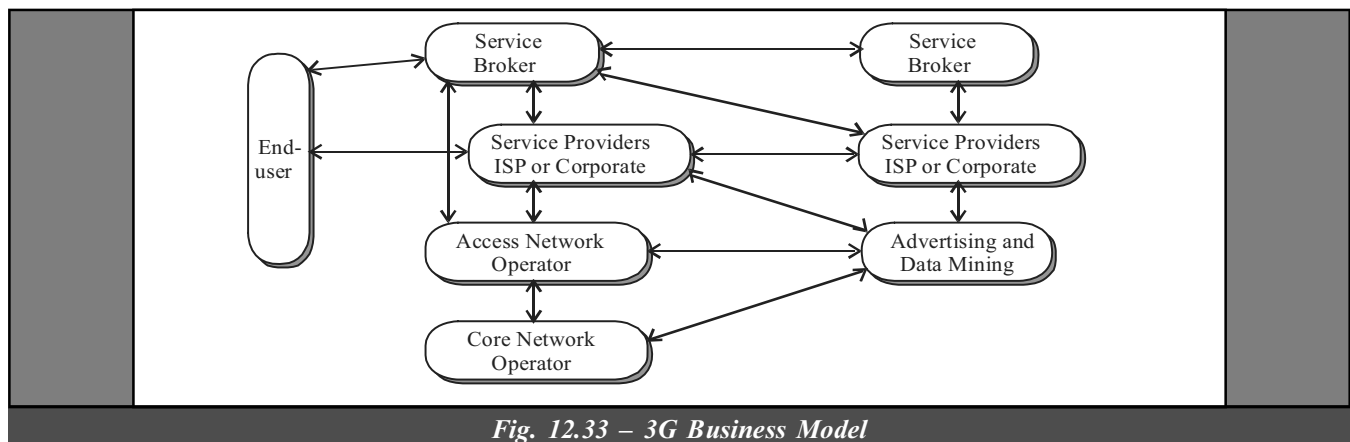


Fig. 12.33 – 3G Business Model

- *Right selection of key applications and other value-added services:* Deployment of key applications and other value-added services, keeping the Indian market segment in view, is extremely important to make business viable.
- *Business model:* The detailed working of a 3G-business model, keeping in view the business opportunities available in India, is extremely important. They should not straight away adopt the business model worked out for other countries. Fine-tuning of the model, keeping in view the Indian market segments/conditions and opportunities, is required. For this, detailed market research and guidance from well-known experts is a must. A business model should be made available well in advance before taking any action on 3G project implementation. A good business plan will help in directing right marketing and network implementation strategies, coupled with adequate sources of funds to be made available at the right time. The whole process is going to be extremely complex and time-consuming. The business model should target the mass market in India and should be viable in a multi-operator environment.
- *Customer relation management (CRM):* 3G will call for a total face uplift of CRM related activities. Failure to take the quality issue into account can be a major contributor to churn a problem. Success of enormous 3G applications and services will call for steps to improve brand recognition, reduced rate plan complexity, negligible fluctuations in the quality of service to take care of negative impact of subsidized handsets and high-price levels, etc. A systematic approach is required to focus on the network operators or service providers and most valuable customers for maximising returns on investment.

Key strategies may include:

- Incentives and tariff bundles that target the customers.
- Loyalty schemes which are appropriate to the value of the business saved.
- Acquisition strategies that attract the right type of customers and aid in reducing fraud and bad debt.
- Proactive customer service.
- A commitment to quality in all aspects of the business.
- Automation to allow network operators and service providers to spend the maximum time in retaining customers.

It is essential for operators to understand their customer needs, but the pace of change in the industry is not making it easy. In the age of GPRS, UTMS, new service choices and increasing competition, the operators need to manage relationships with customers differently.

- *Marketing strategy:* Marketing should work for right value-added services, look for increased national and international roaming and competitive pricing. Operators should try to add customers as rapidly as possible to utilise their high capacity 3G networks. They also have to concentrate on consumer brands for their services as well as establishing partnerships with content providers to make their data offerings far more diverse and differentiated. Marketing should have a clear idea what customers will be willing to buy and at what price. The old concept of – “who owns the customer” – will become – “who the customer owns”.
- *Distribution channel:* One of the potential means of distributing 3G products and services is through the growing ISP industry. Conversely, ISP will regard 3G networks as opening up new distribution channels for their services. The fit between 3G operators, ISP and media companies is seen as a critical success factor for any future 3G business. Food retail giants, banks and supermarkets can be good distribution channels, as they have a greater reach into mass-market segments and their lower commission payments make them far more attractive. New channels for 3G terminals will include car manufacturers, dealerships for consoles and terminals, traditional IT retailers, dealers and system integrators for the distribution of 3G computers to the consumer corporate markets. Since in the beginning, handset prices are going to be high, some sort of subsidy can be thought of. Non-handset terminals need not be subsidised for obvious reasons of their multiple uses and may not generate 3G traffic.
- *Pricing:* On the pricing front, 3G operators may consider pricing of voice and data as two separate services, since most 3G networks will be IP or ATM based and therefore will allow pricing on volume basis, i.e., per megabyte basis rather than per minute basis (this was the case with 2G systems). This will allow charging for data services on a usage based principal, and hence will encourage more usage and in turn more revenue. This will also attract more subscribers to join 3G services.
- *Fraud systems support:* There will be frauds peculiar to the mobile Internet with unpredictable forms. Indeed, it is very likely that the mobile Internet will change the dimension of fraud. All the forms known today on 2G systems will carry a slightly altered form – as mobile Internet fraud call selling – where criminals sell high-value calls at a substantial discount to other people using call-diversion facilities. GPRS and UTMS with extended bandwidth will provide an opportunity of call selling on a much large scale. Similarly, sophisticated variations of premium rate – service fraud, where criminals direct calls to their own premium rate numbers to generate revenue are likely to continue in the mobile Internet sphere. A foolproof, fraud systems support needs to be devised to check frauds of various kinds and save operators from incurring heavy losses on account of fraudulent customers.

#### (viii) Need for Partnership

The relationship between content developers, software developers, platform partners, application developers and mobile network operators will have to become more integrated as the 3G value chain becomes increasingly complex. No single provider can supply across the supply chain to each segment. This new market will need cooperation and partnership across the value chain, if the 3G opportunity is to be completely realised. Indian cellular operators, etc. need to come to a common understanding of establishing a mutually agreed upon “3G applications/services mass education programme” across the country. However, the marketing and sales strategy followed by them can be totally different from each other.

**(ix) Steps in the Right Direction**

In the “National Frequency Allocation Plan – 2000” (NFAP-2000), Ministry of Communications (Wireless Planning and Coordination Wing) has indicated the following:

- Catering to additional requirements may be coordinated in the frequency band of 902.5-915 MHz, paired with 947.5-960 MHz.
- Requirement of cellular and WLL services in the frequency band 1700-2000 MHz may be coordinated on a case-to-case basis, initially (10+10) MHz in the frequency band of 1710-1785 MHz, paired with 1805-1880 MHz. Additional (10+10) MHz may also be co-ordinated on a case-by-case basis, subsequently in the frequency band of 1710-1785 MHz, paired with 1805-1880 MHz.
- Requirements of IMT-2000 in the frequency band of 1885-2025 MHz, paired with 2110-2200 MHz, will be considered and appropriately coordinated with existing users on priority basis. IMT-2000 requirements for extension bands can also be considered in future.
- Induction of cellular/WLL technologies, especially in the rural areas, may be considered for coordination on case-by-case basis, in the frequency band of 450.5-457.5 MHz, paired with 460.5-467.5 MHz.

Detailed study of NFAP-2000 clearly indicates that the Government of India is determined to give a helping hand to the growth of futuristic technologies in order to allow economic growth and improve the quality of life of our country.

With the opening of domestic long distance (DLD), 100% investment by foreign companies in ISP, availability of greater bandwidth for the IT industry, dismantling of DoT and the determination of the government to end VSNL's monopoly in the long run are some of the positive indications of the explosive growth of futuristic technology related applications.

**(x) The Portability Quest**

Wireless communication is driving the development of new standards to bring about dramatic changes in the enterprise communication landscape.

**(xi) Highlights**

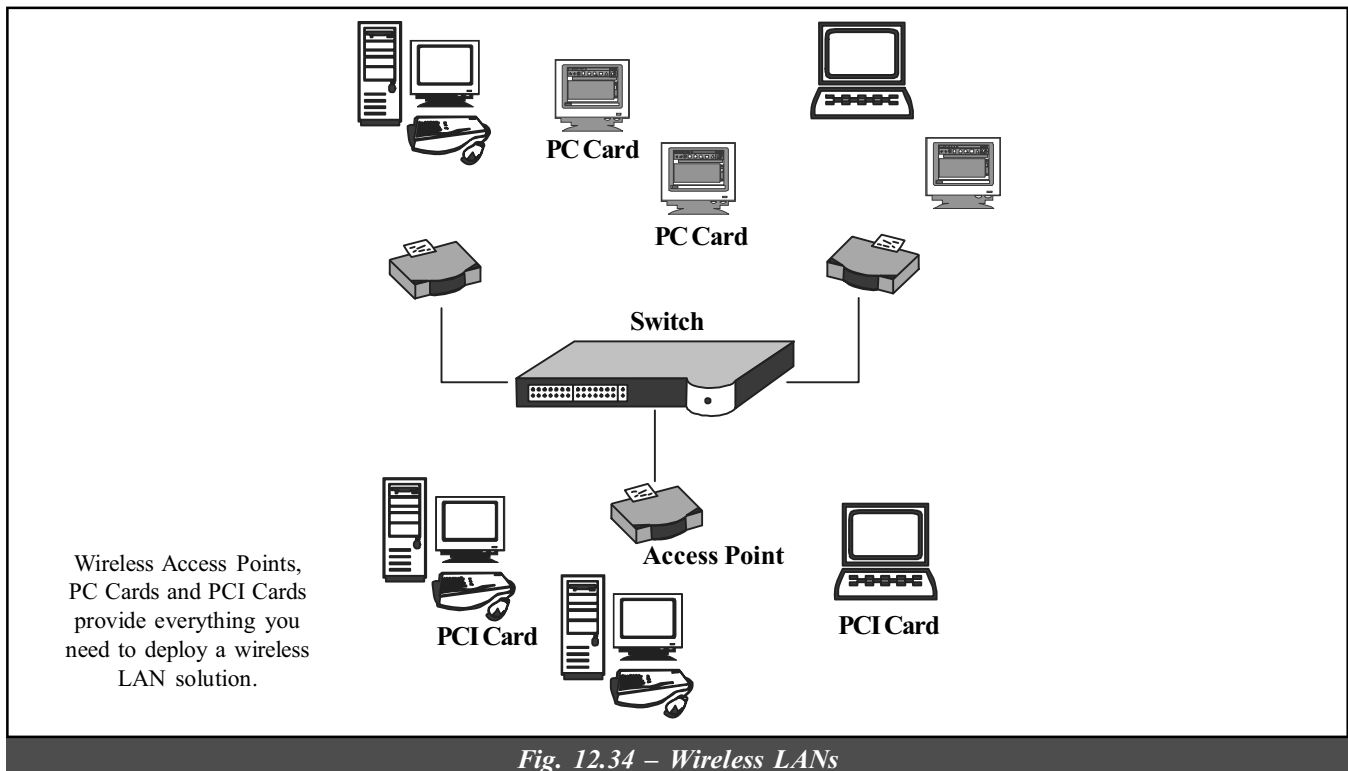
- Wireless LAN is not a replacement for the wired infrastructure, but a significant complement to what exists.
- Wireless LAN will soon move beyond the domain of data computing on mobile device to voice networking.
- *Wireless technologies in the PAN, LAN, WAN and line-of-sight space are set to proliferate during the next two to three years.*

**Wireless LANs**

Wiring an office or buildings for a LAN is often difficult and costly. Older buildings do not have conduits for coaxial cables or for additional twisted-pair wire and the conduits in newer buildings may not have enough room to pull additional wiring through. Repairing mistakes and damages to wiring is cumbersome and costly, as are major relocations of work stations and other components. One solution to such problems is installing a ‘wireless LAN’, using one of several wireless technologies, e.g., LAN Radio, which may involve a high frequency radio technology similar to digital cellular or a low frequency radio technology called *spread spectrum*. The other wireless LAN technology is called infrared because it uses beams of infrared light to establish network links between LAN components. A wireless LAN eliminates or greatly reduces the need for wires and cables, thus making a LAN easier to set up, relocate and maintain. However, current wireless technologies have higher initial costs and other limitations, e.g., infrared LAN transmits faster than radio LANs but is limited to the line-of-sight arrangements to a maximum of about 80 feet between components.

Wireless LANs have been around for years, but their uptake has been slow. The growth has been hampered by a lack of adequate standards and many users feel more comfortable with wired networks. However, Wireless LAN gear will gain a foothold in small companies and branch offices because the devices are portable. Standardisation and an emphasis on making the devices easier to install should boost their popularity. In addition, the wireless devices linking workers to supply chain applications will be a driving force, depending upon the processes of the firm.

Currently, Wireless LAN is not a replacement for the wired infrastructure, but a significant complement to what exists. The schools, manufacturing companies, hospitals and offices purchase their wireless LAN systems for two predominant reasons – to seek an increase in user productivity and IT team productivity. This translates to “real-time access to information” while at work, mobility and reduced cost of ownership.



### The Technology

Most Wireless LANs use spread spectrum radio transmission techniques. The spread spectrum technology was first introduced with the objective of improving both message integrity and security. Spread spectrum systems are designed to be resistant to noise, interference, jamming and unauthorised detection. The spread spectrum transmitters send their signals out over a broad range of frequencies at a very low power, in contrast to the narrow band radios that concentrate their entire power into a single frequency. There are several ways to implement spread spectrum transmission, the two most common being Direct Sequence (DS) and Frequency Hopping (FH).

Both the techniques present unintended receivers with a difficult problem. In the case of DS, an eavesdropper must know the chipping (spreading) code. Anybody trying to intercept a FH transmission must understand the hopping pattern. In both cases, the specific frequency bands (or portion thereof in the case of DS) and modulation techniques in use must also be known. Radio systems also use a form of data scrambling for purely technical reasons, which is to assist in managing the timing and decoding of radio signals. An intended receiver would also need to know this scrambling pattern.

Infrared-based Wireless LANs are often used in high-security applications because infrared signals do not penetrate solid objects like walls. But all of these techniques allow the use of encryption, and indeed, many wireless LAN products include encryption features as a standard or optional component. The IEEE 802.11 standard, for example, includes a security technique known as Wired Equivalent Privacy (WEP), which is based on the use of 64-bit keys and the popular RC4 encryption algorithm. Users without knowledge of the current key (password) will find themselves excluded from network traffic. Encryption, as noted above, is always advisable on any network and is certainly easier to implement in Wireless LANs than in their wired counterparts.

Most Wireless LAN products have the ability of an authentication management function, to specifically authorise or exclude individual wireless stations. Thus, an individual wireless user can be included in a network or locked out at any time. The users may also need to know a wide variety of information, including radio domains, channels (specific frequencies or hopping patterns) and sub-channels.

### The M-commerce Matrix

The WAP Client is a micro-browser equipped device through which information can be accessed. It may be a mobile phone, a personal digital assistant (PDA) or a pager. The leaders in this segment, Nokia, Ericsson, Motorola and Siemens have all-embraced WAP and have introduced WAP enabled mobile devices. Nokia, the industry leader in mobile phone

production, is developing pen-based mobile computer phones with Palm Computing, the leader in Palmtops, with touch-sensitive screens instead of keyboards. All these mobile devices come with an in-built microbrowser like Microsoft's Microsoft Mobile Explorer.

The WAP Gateway/Server serves as the interface between the two sets of network protocols – Wireless (WAP) and Wireline or Internet Protocol (TCP/IP), converting requests and information from Wireless Markup Language (WML) to HTML and *vice versa*. A WAP gateway/server is in between the wireless carrier's network on one side and the Internet or corporate Intranet on the other. Gateways can be located within the carrier's or corporate's firewalls or both. The fast pace at which the conventional websites are being WAP-enabled basically means that the content will soon be available in WML itself. So, the WAP gateway/server will only serve as an interface between two stacks of the protocol, and receive and send the information to a WAP client in compact binary form. WAP gateways/servers are available from a number of organisations like Ericsson, Nokia, Phone.com, ApiON, etc. In India, organisations like Satyam Infoway, Integra Microsystems and Silicon Automation have their own versions of WAP gateways.

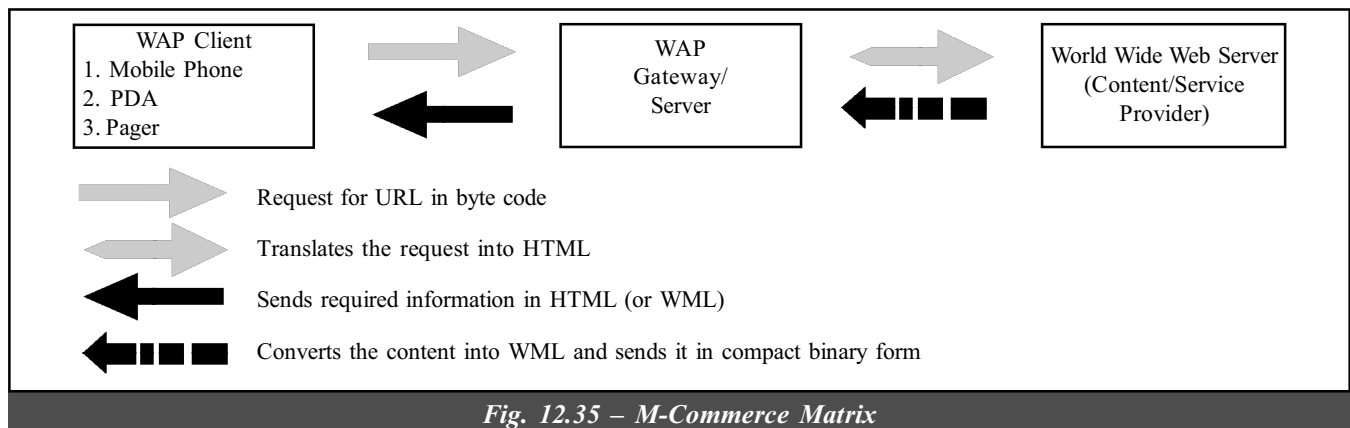


Fig. 12.35 – M-Commerce Matrix

## THE LEAST YOU NEED TO KNOW

- *What is Wireless Application Protocol (WAP)? An understanding of handheld devices and problems?*
- *Relationship with Mobility and Commerce.*
- *What is Bluetooth? What is the security of WAP transactions?*
- *What is the expansion of mobile computing in four dimensions?*
- *An understanding of wireless web, web security, PKI, Alias card, positional services and WAP as the future of technology?*
- *What is the progress of WAP in India?*
- *What is Mobile computing? What are its stages?*

## WRAP UP

- The Wireless Application Protocol (WAP) is a set of specifications, developed by the WAP Forum that lets developers using wireless markup language (WML) build networked applications designed for handheld wireless devices. The WAP was designed to make it easier to create networked applications for handheld devices despite the drawbacks.
- Handheld devices are more limited than desktop computers in several important ways. A handheld's constraints mean that it is usually impossible to directly port desktop applications to a wireless handheld device. WAP is the gateway to a network of mobile data.
- Soon, electronic office devices will communicate with each other over short distances without wires. Bluetooth is the new wireless communication technology which will eliminate cable chaos and pave the way for mobile business.

- The expansion of mobile computing is in four dimensions:
  - Wireless delivery technology and switching methods.
  - Mobile information access devices.
  - Mobile data internet working standards and equipment.
  - Mobile computing based business applications.
- The wireless internet allows full access to the web via a mobile handset or any mobile device such as a full organiser. The bundle of technologies developed under the control of Internet Engineering Task Force (IETF) is known as Public Key Infrastructure (PKI). Some people see the wireless internet as the way of the future and is pushing hard to develop a couple of related technologies like Bluetooth, UMTS, etc.
- WAP is a protocol that supports mobile computing. It creates a parallel Internet and limits the content that the end-users are able to get. Security issues in mobile computing include Authentication, Identification, Secure Messaging, Certification, etc. 3G will make possible most of the personal communications, be it in the form of phone calls, data images or live video.
- Wireless LANs have been around for years, but their uptake has been slow. The growth has been hampered by a lack of adequate standards and many users feel more comfortable with wired networks. Wireless LAN is not a replacement for the wired infrastructure, but a significant complement to what exists.

### **TAKING STOCK**

1. Define WAP. What are the handheld devices? What are the problems with handheld devices?
2. 'Wireless Application Protocol is the gateway to a new world of mobile data.' Comment.
3. Some of the wireless technologies comment WAP as 'Wasteful Application protocol!'. Why should they call it like that?
4. What do you mean by bluetooth? Which are the four dimensions found in the expansion of mobile computing?
5. 'Every security system developed by the human mind can be broken by another human mind.' What are your views?
6. 'The wireless web is the consolidation of peoples' aspiration and craving for information to be available anywhere any time.' Justify this statement.
7. What is the scope for WAP in the Indian market? Comment on its potential.
8. What do you mean by mobile computing? Discuss the stages of mobile computing.
9. 'There is no Killer Application', instead, technology is helping to create a 'Killer Environment' in the case of third generation mobile'. Discuss.
10. Explain the 3G business mobile and discuss on '3G' evolution.
11. What are wireless LANs? What is the technology involved in this?